PPMI-PEG mQTL / metQTL Overlap Analysis

Longitudinal mQTL meta-hits overlapped with PEG1 metabolomics QTLs, annotated via Ensembl

August 12, 2026

Overview

This report combines four analysis stages:

1. **metQTL FDR handling and overlap** — combining PEG1 c18 + hilic trans-metQTL results, and overlapping FDR-significant SNP-metabolite pairs against the classified longitudinal PPMI mQTL results (from the meta-analysis hit list, re-tested with a mixed model across follow-up visits).
2. **Gene annotation** — resolving the overlapping SNPs to nearest/overlapping genes via the Ensembl REST API, since a shared SNP driving both a methylation and a metabolite association is the strongest signal of a pleiotropic locus worth following up.
3. **PD GWAS overlap and union table** — checking whether the case meta-analysis mQTL hits fall near a known PD risk locus from Nalls et al. (2019, Table S2), then building a union table of case hits supported by *either* the metQTL overlap *or* GWAS proximity (or both).
4. **Metabolite compound annotation** — resolving as many of the overlap metabolite features as possible from their raw `mz_rt` labels to putative compound identities, via xmsannotator mass/retention-time matching against HMDB, KEGG, and LipidMaps (see Step 6).

Metabolite features (`C18_mz_rt_... / HILIC_mz_rt_...`) are carried under their original mass-spec feature labels through Steps 1-5. No collaborator compound-ID mapping table ever materialized, so Step 6 instead resolves them independently via xmsannotator mass/retention-time matching — with partial coverage (23/33 features get any match, 11/33 a high-confidence named compound; see Step 6 for caveats).

A methodological note carried through this analysis: the `trans_metqtls.txt.gz` files are generated with a liberal `pvOutputThreshold` (1e-2, set in `pipelines/PEG_METAB/config.yaml`) so that enough candidate SNPs survive to make the overlap with the longitudinal mQTL hits informative. The complete output is written gzipped and kept intact; the truncation to the first 1e+06 rows now happens **at read time** here, by streaming the gzip and stopping at the cap, rather than by a manual `head` on the cluster before transfer.

Two consequences follow, and both matter:

- The rows we load are *not* the full test space, so a BH correction recomputed on them alone would be invalid — it would rank an already-preselected low-p subset against itself instead of against the true, much larger, denominator of all tests performed. We therefore use **MatrixEQTL's own per-file FDR column**, which is computed against that true denominator before any truncation, rather than recomputing it here. This requires `noFDRsaveMemory = FALSE in run_metqtl.R` ; with it on, MatrixEQTL omits the FDR column entirely.
- The cap is only safe because MatrixEQTL sorts its output by p-value ascending when it computes FDR. Truncating therefore discards the least significant tail, not an arbitrary slice.

c18 and hilic are treated as separate multiple-testing families rather than one merged correction, since a combined correction isn't recoverable from truncated data anyway.

Step 1: Combine metQTL results and apply FDR filtering

```
combine_and_fdr <- function(group_label, files, fdr_threshold = 0.05) {

  combined <- bind_rows(lapply(files, function(f) {
    # read_tsv streams the gzip rather than expanding it, and n_max stops the
    # read at the cap, so peak memory is bounded by the cap and not by the
    # on-disk file — which at pval_trans = 1e-2 is several GB compressed.
    read_tsv(f, col_types = cols(SNP = col_character(), gene = col_character(),
                                 beta = col_double(), `t-stat` = col_double(),
                                 `p-value` = col_double(), FDR = col_double()),
             n_max = MAX_TRANS_ROWS, guess_max = Inf) %>%
      mutate(source_file = basename(dirname(f)))
  }))

  # Safety net for the row cap. Rows are p-value ascending, so significant rows
  # can only be lost if a file both hit the cap AND was still significant at the
  # last retained row. Anything else means the cap only cut dead tail.
  cap_check <- combined %>%
    group_by(source_file) %>%
    summarise(n = n(), worst_fdr = max(FDR, na.rm = TRUE), .groups = "drop") %>%
    mutate(hit_cap = n >= MAX_TRANS_ROWS,
           clipped_signal = hit_cap & worst_fdr < fdr_threshold)
  print(cap_check)
  if (any(cap_check$clipped_signal)) {
    cat(sprintf(paste0("\n  *** MAX_TRANS_ROWS (%s) CLIPPED SIGNIFICANT ROWS in: %s. ",
                       "The FDR<%.2f set is truncated — raise the cap and re-run. ***\n"),
                format(MAX_TRANS_ROWS, big.mark = ","),
                paste(cap_check$source_file[cap_check$clipped_signal], collapse = ", "),
                fdr_threshold))
  }

  sig <- combined %>% filter(FDR < fdr_threshold)

  # The combined table is the capped read in full and is never read back — only
  # `sig` feeds the overlap. write_csv gzips on a .gz extension, which takes it
  # from ~2.4 GB to a fraction of that per group. `sig` stays plain: it is small
  # and gets opened by hand.
  out_all <- file.path(RESULTS_DIR, sprintf("peg1_%s_metqtl_combined_fdr.csv.gz", group_label))
  out_sig <- file.path(RESULTS_DIR, sprintf("peg1_%s_metqtl_sig_fdr05.csv", group_label))
  write_csv(combined, out_all)
  write_csv(sig, out_sig)

  list(combined = combined, sig = sig)
}

cases_metqtl    <- combine_and_fdr(
  "cases",
  c("../../results/PEG_METAB/metqtl/c18_peg1_cases/trans_metqtls.txt.gz",
    "../../results/PEG_METAB/metqtl/hilic_peg1_cases/trans_metqtls.txt.gz")
)
```

```
## # A tibble: 2 × 5
##   source_file            n worst_fdr hit_cap clipped_signal
##   <chr>              <int>     <dbl> <lgl>   <lgl>
## 1 c18_peg1_cases   1000000     0.583 TRUE    FALSE
## 2 hilic_peg1_cases 1000000     0.636 TRUE    FALSE
```

```
controls_metqtl <- combine_and_fdr(
  "controls",
  c("../../results/PEG_METAB/metqtl/c18_peg1_controls/trans_metqtls.txt.gz",
    "../../results/PEG_METAB/metqtl/hilic_peg1_controls/trans_metqtls.txt.gz")
)
```

```
## # A tibble: 2 × 5
##   source_file               n worst_fdr hit_cap clipped_signal
##   <chr>                 <int>     <dbl> <lgl>   <lgl>
## 1 c18_peg1_controls   1000000     0.575 TRUE    FALSE
## 2 hilic_peg1_controls 1000000     0.545 TRUE    FALSE
```

```
metqtl_cases_sig    <- cases_metqtl$sig
metqtl_controls_sig <- controls_metqtl$sig
```

```
metqtl_summary <- tibble(
  group           = c("Cases", "Controls"),
  total_tests     = c(nrow(cases_metqtl$combined), nrow(controls_metqtl$combined)),
  fdr_sig_pairs   = c(nrow(metqtl_cases_sig), nrow(metqtl_controls_sig))
)
kable(metqtl_summary, caption = "PEG1 metQTL: total tests combined (c18+hilic) and FDR<0.05 pairs")
```

PEG1 metQTL: total tests combined (c18+hilic) and FDR<0.05 pairs

group	total_tests	fdr_sig_pairs
Cases	2e+06	11333
Controls	2e+06	18036

Step 2: Overlap with classified longitudinal mQTL results

Overlap is defined by shared SNP identity, cases-vs-cases and controls-vs-controls kept separate (different populations, not directly comparable). A single pleiotropic SNP can legitimately show up with multiple significant CpGs *and* multiple significant metabolites — that many-to-many structure is expected biology, not a join error.

```
overlap_group <- function(group_label, mqtl_classified_file, metqtl_sig) {

  mqtl <- read_csv(mqtl_classified_file,
                   col_types = cols(pair_id = col_character(), snp_id = col_character(),
                                    cpg_id = col_character(), status = col_character(),
                                    warnings = col_character(), category = col_character(),
                                    same_sign_as_meta = col_logical(), replicated = col_logical(),
                                    dynamic = col_logical(), .default = col_double()),
                   guess_max = Inf)

  overlap <- mqtl %>%
    inner_join(metqtl_sig, by = c("snp_id" = "SNP"), relationship = "many-to-many") %>%
    rename(metabolite_feature = gene, metqtl_beta = beta, metqtl_pval = `p-value`, metqtl_fdr = FDR)

  write_csv(overlap, file.path(RESULTS_DIR, sprintf("mqtl_metqtl_overlap_%s.csv", group_label)))

  # Carry the mQTL denominator with the result so the summary table can report
  # it instead of a hardcoded literal that goes stale on the next re-run.
  attr(overlap, "n_mqtl_pairs") <- nrow(mqtl)
  overlap
}

overlap_cases <- overlap_group("cases",
  "../../results/PPMI_LONG/longitudinal_mqtl_cases_classified.csv", metqtl_cases_sig)

overlap_controls <- overlap_group("controls",
  "../../results/PPMI_LONG/longitudinal_mqtl_controls_classified.csv", metqtl_controls_sig)
```

```
overlap_summary <- tibble(
  group                = c("Cases", "Controls"),
  # Derived, not hardcoded: these were literals (18986, 15282) from a run two
  # hit-lists ago and silently misreported the denominator once PPMI_LONG was
  # re-run against a new meta-analysis.
  mqtl_pairs_tested     = c(attr(overlap_cases, "n_mqtl_pairs"),
                            attr(overlap_controls, "n_mqtl_pairs")),
  metqtl_sig_pairs       = c(nrow(metqtl_cases_sig), nrow(metqtl_controls_sig)),
  unique_shared_snps     = c(n_distinct(overlap_cases$snp_id), n_distinct(overlap_controls$snp_id)),
  overlap_rows           = c(nrow(overlap_cases), nrow(overlap_controls))
)
kable(overlap_summary, caption = "mQTL x metQTL overlap summary")
```

mQTL x metQTL overlap summary

group	mqtl_pairs_tested	metqtl_sig_pairs	unique_shared_snps	overlap_rows
Cases	22407	11333	32	74
Controls	4470	18036	8	13

Longitudinal category breakdown among overlapping SNPs

```
cat_cases <- overlap_cases %>% distinct(snp_id, category) %>% count(category) %>% mutate(group = "Cases")
cat_controls <- overlap_controls %>% distinct(snp_id, category) %>% count(category) %>% mutate(group = "Controls")
bind_rows(cat_cases, cat_controls) %>%
  pivot_wider(names_from = group, values_from = n, values_fill = 0) %>%
  kable(caption = "Longitudinal classification of SNPs shared with metQTL hits")
```

Longitudinal classification of SNPs shared with metQTL hits

category	Cases	Controls
replicated_static	32	8

Dynamic mQTLs (cases): metQTL overlap status

The overlap above is an *inner join* — it only shows hits that overlapped a metQTL, so a longitudinally “dynamic” mQTL with no metabolite signal would be invisible by construction. To see all dynamic hits regardless of metQTL status, we take the classified longitudinal results (cases) as the **left** table and left-join the metQTL significant set, keeping every dynamic hit whether or not it matched.

```
cases_classified_full <- read_csv("../../results/PPMI_LONG/longitudinal_mqtl_cases_classified.csv",
                                   col_types = cols(pair_id = col_character(), snp_id = col_character(),
                                                    cpg_id = col_character(), status = col_character(),
                                                    warnings = col_character(), category = col_character(),
                                                    same_sign_as_meta = col_logical(), replicated = col_logical
(),
                                                    dynamic = col_logical(), .default = col_double()),
                                   guess_max = Inf)

dynamic_hits <- cases_classified_full %>%
  filter(category == "replicated_and_dynamic") %>%
  left_join(metqtl_cases_sig %>% select(SNP, gene, `p-value`, FDR),
            by = c("snp_id" = "SNP"), relationship = "many-to-many") %>%
  rename(metabolite_feature = gene, metqtl_pval = `p-value`, metqtl_fdr = FDR) %>%
  mutate(has_metqtl_match = !is.na(metabolite_feature))

kable(dynamic_hits %>%
        select(snp_id, cpg_id, main_beta, main_fdr, interaction_p, interaction_fdr,
               has_metqtl_match, metabolite_feature, metqtl_fdr),
      caption = "All 'replicated_and_dynamic' case mQTLs, with metQTL match status (left join — unmatched rows sho
w NA)")
```

All ‘replicated_and_dynamic’ case mQTLs, with metQTL match status (left join — unmatched rows show NA)

snp_id	cpg_id	main_beta	main_fdr	interaction_p	interaction_fdr	has_metqtl_match	metabolite_feature	metqtl_fdr
4:184324768	cg27037990	0.0317113	0.000916	1e-07	0.0014908	FALSE	NA	NA

```
no_metqtl <- dynamic_hits %>% filter(!has_metqtl_match)
cat("Dynamic mQTLs (cases) with NO metQTL overlap:", n_distinct(no_metqtl$snp_id), "\n")
```

```
## Dynamic mQTLs (cases) with NO metQTL overlap: 1
```

```
kable(no_metqtl %>% select(snp_id, cpg_id, main_beta, main_fdr, interaction_p, interaction_fdr),
      caption = "Dynamic mQTLs (cases) with no corresponding metabolite QTL signal")
```

Dynamic mQTLs (cases) with no corresponding metabolite QTL signal

snp_id	cpg_id	main_beta	main_fdr	interaction_p	interaction_fdr
4:184324768	cg27037990	0.0317113	0.000916	1e-07	0.0014908

Step 3: Gene annotation via Ensembl REST API

```
unique_snps <- union(overlap_cases$snp_id, overlap_controls$snp_id)

WINDOW <- 250000
BASE_URL <- "https://rest.ensembl.org"

annotate_snp <- function(snp_id) {
  parts <- strsplit(snp_id, ":")[[1]]
  chr <- parts[1]
  pos <- as.integer(parts[2])

  query_region <- function(start, end) {
    url <- sprintf("%s/overlap/region/human/%s:%d-%d?feature=gene;content-type=application/json",
                   BASE_URL, chr, start, end)
    resp <- tryCatch(GET(url), error = function(e) NULL)
    if (is.null(resp) || status_code(resp) != 200) return(NULL)
    fromJSON(content(resp, as = "text", encoding = "UTF-8"), flatten = TRUE)
  }

  hit <- query_region(pos, pos)
  Sys.sleep(0.1)

  if (!is.null(hit) && length(hit) > 0 && is.data.frame(hit) && nrow(hit) > 0) {
    gene_labels <- ifelse(is.na(hit$external_name) | hit$external_name == "",
                          hit$id, hit$external_name)
    genes <- unique(gene_labels)
    return(tibble(snp_id = snp_id, gene = paste(genes, collapse = ";"),
                  relationship = "overlapping", distance_bp = 0))
  }

  hit_wide <- query_region(max(1, pos - WINDOW), pos + WINDOW)
  Sys.sleep(0.1)

  if (is.null(hit_wide) || length(hit_wide) == 0 || !is.data.frame(hit_wide) || nrow(hit_wide) == 0) {
    return(tibble(snp_id = snp_id, gene = NA_character_,
                  relationship = paste0("none_within_", WINDOW, "bp"), distance_bp = NA_integer_))
  }

  hit_wide <- hit_wide %>%
    mutate(dist = pmax(0, pmax(start - pos, pos - end)),
           gene_label = ifelse(is.na(external_name) | external_name == "", id, external_name))

  nearest <- hit_wide %>% slice_min(dist, n = 1, with_ties = FALSE)

  tibble(snp_id = snp_id, gene = nearest$gene_label,
         relationship = "nearest", distance_bp = nearest$dist)
}

annotations <- map_dfr(unique_snps, function(s) {
  tryCatch(annotate_snp(s), error = function(e) {
    tibble(snp_id = s, gene = NA_character_, relationship = "query_failed", distance_bp = NA_integer_)
  })
})

write_csv(annotations, file.path(RESULTS_DIR, "overlap_snp_gene_annotations.csv"))

overlap_cases_annot <- overlap_cases %>%
  left_join(annotations, by = "snp_id") %>%
  rename(nearest_gene = gene, gene_relationship = relationship, gene_distance_bp = distance_bp)

overlap_controls_annot <- overlap_controls %>%
  left_join(annotations, by = "snp_id") %>%
  rename(nearest_gene = gene, gene_relationship = relationship, gene_distance_bp = distance_bp)

write_csv(overlap_cases_annot, file.path(RESULTS_DIR, "mqtl_metqtl_overlap_cases_annotated.csv"))
write_csv(overlap_controls_annot, file.path(RESULTS_DIR, "mqtl_metqtl_overlap_controls_annotated.csv"))
```

Annotation outcome

```
kable(as.data.frame(table(relationship = annotations$relationship)),
      caption = "SNP -> gene annotation outcome (overlapping gene body, nearest within 250kb, or none found)")
```

SNP -> gene annotation outcome (overlapping gene body, nearest within 250kb, or none found)

relationship	Freq
nearest	12
none_within_250000bp	14
overlapping	14

Genes implicated, cases

```
genes_cases <- overlap_cases_annot %>%
  distinct(snp_id, nearest_gene) %>%
  filter(!is.na(nearest_gene), nearest_gene != "") %>%
  count(nearest_gene, sort = TRUE)

kable(genes_cases, caption = "Unique genes implicated by shared SNPs — Cases")
```

Unique genes implicated by shared SNPs — Cases

nearest_gene	n
ERLIN1	16
SPCS2P2	3
MYT1L	2
AGPS	1
CYTH1	1
ENSG00000248752	1

Genes implicated, controls

```
genes_controls <- overlap_controls_annot %>%
  distinct(snp_id, nearest_gene) %>%
  filter(!is.na(nearest_gene), nearest_gene != "") %>%
  count(nearest_gene, sort = TRUE)

kable(genes_controls, caption = "Unique genes implicated by shared SNPs — Controls")
```

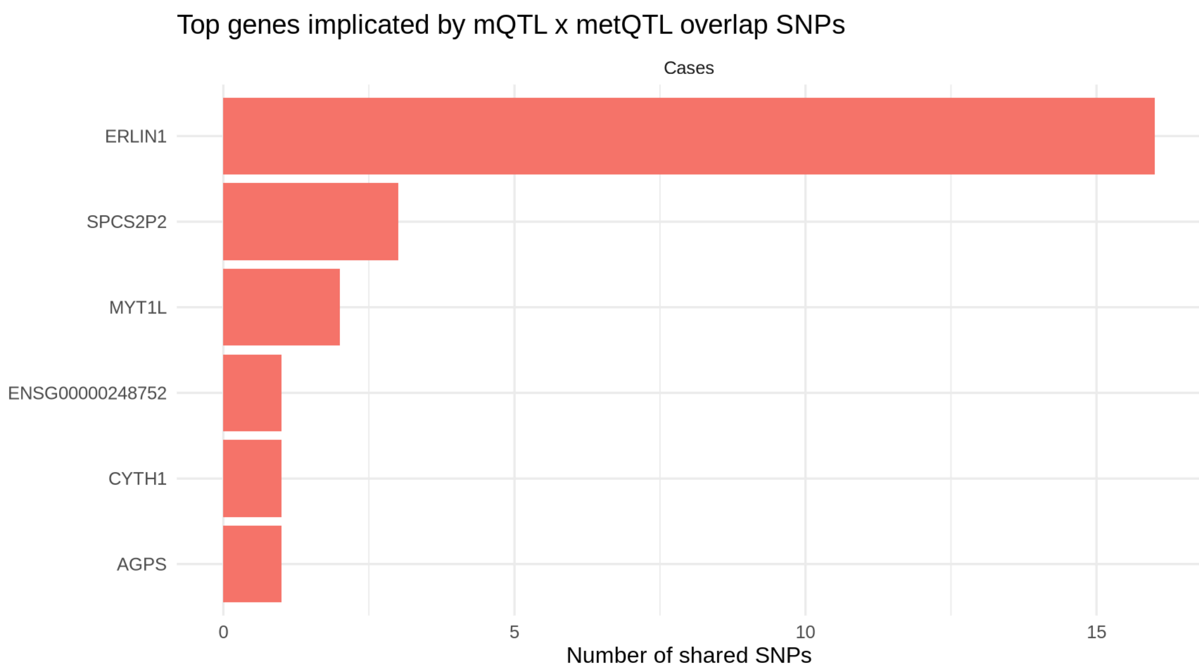
Unique genes implicated by shared SNPs — Controls

nearest_gene	n

Gene frequency, top hits

```
top_genes <- bind_rows(
  genes_cases %>% mutate(group = "Cases"),
  genes_controls %>% mutate(group = "Controls")
) %>%
  group_by(group) %>%
  slice_max(n, n = 10) %>%
  ungroup()

ggplot(top_genes, aes(x = reorder(nearest_gene, n), y = n, fill = group)) +
  geom_col(show.legend = FALSE) +
  coord_flip() +
  facet_wrap(~group, scales = "free") +
  labs(x = NULL, y = "Number of shared SNPs",
       title = "Top genes implicated by mQTL x metQTL overlap SNPs") +
  theme_minimal(base_size = 12)
```



Step 4: Overlap with Nalls et al. PD GWAS loci

Using the curated Nalls et al. (2019) locus table (Table S2) — already restricted to genome-wide significant, LD-independent lead SNPs — rather than the looser nominal-significance (p<0.05) SNP list, since the latter would flag a large fraction of any SNP set as a "GWAS hit" by chance alone.

Because these are independent lead SNPs from a clumping procedure, they are unlikely to be the exact variant present in the PPMI/PEG genotype panel. We therefore use a **window-based match (±250kb)** rather than requiring identical `chr:pos` — the same logic used for the Ensembl gene-annotation step above. This window is a judgment call; tighten or loosen `GWAS_WINDOW` below as appropriate.

```r
GWAS_WINDOW <- 250000   # bp; matches the Ensembl gene-annotation window

nalls_raw <- read_tsv("../../Downloads/reference/Nalls_GWAS/PDNallsP005_TableS2.csv",
                       col_types = cols(.default = col_character()), guess_max = Inf)

qtl_gene_col   <- grep("QTL_Nominated_Gene", names(nalls_raw), value = TRUE)[1]
p_all_col      <- grep("^P_all_studies$", names(nalls_raw), value = TRUE)[1]
locus_num_col  <- grep("Locus_Number", names(nalls_raw), value = TRUE)[1]

nalls <- nalls_raw %>%
  transmute(
    nalls_snp     = SNP,
    CHR           = as.integer(CHR),
    BP            = as.integer(gsub(",", "", BP)),
    nearest_gene  = Nearest_Gene,
    qtl_gene      = .data[[qtl_gene_col]],
    p_all_studies = as.numeric(.data[[p_all_col]]),
    locus_number  = as.integer(.data[[locus_num_col]])
  )

stopifnot("Nalls BP parsing failed" = sum(is.na(nalls$BP)) == 0)
stopifnot("Nalls CHR parsing failed" = sum(is.na(nalls$CHR)) == 0)

cat("Nalls loci loaded:", nrow(nalls), "\n")
```

```
## Nalls loci loaded: 90
```

```r
cases_meta <- read_csv("../../results/PPMI_LONG/longitudinal_mqtl_cases_classified.csv",
                        col_types = cols(pair_id = col_character(), snp_id = col_character(),
                                         cpg_id = col_character(), status = col_character(),
                                         warnings = col_character(), category = col_character(),
                                         same_sign_as_meta = col_logical(), replicated = col_logical(),
                                         dynamic = col_logical(), .default = col_double()),
                        guess_max = Inf)

case_snps <- cases_meta %>%
  distinct(snp_id) %>%
  mutate(chr = str_extract(snp_id, "^[^:]+"),
         pos = as.integer(str_extract(snp_id, "(?<=:)[0-9]+")))

match_gwas <- function(chr, pos) {
  cand <- nalls %>% filter(as.character(CHR) == chr)
  if (nrow(cand) == 0) {
    return(tibble(is_gwas_hit = FALSE, gwas_dist_bp = NA_integer_,
                  gwas_nearest_gene = NA_character_, gwas_locus_number = NA_integer_))
  }
  dists <- abs(cand$BP - pos)
  i <- which.min(dists)
  tibble(is_gwas_hit = dists[i] <= GWAS_WINDOW, gwas_dist_bp = dists[i],
         gwas_nearest_gene = cand$nearest_gene[i], gwas_locus_number = cand$locus_number[i])
}

gwas_matches <- case_snps %>%
  mutate(m = pmap(list(chr, pos), match_gwas)) %>%
  unnest(m)

cat("Unique case-hit SNPs tested against Nalls loci:", nrow(gwas_matches), "\n")
```

```
## Unique case-hit SNPs tested against Nalls loci: 19282
```

```r
cat("Within", GWAS_WINDOW/1000, "kb of a known PD GWAS locus:", sum(gwas_matches$is_gwas_hit), "\n")
```

```
## Within 250 kb of a known PD GWAS locus: 905
```

GWAS-adjacent case hits

```
gwas_hits_table <- gwas_matches %>%
  filter(is_gwas_hit) %>%
  arrange(gwas_dist_bp) %>%
  select(snp_id, gwas_nearest_gene, gwas_locus_number, gwas_dist_bp)

kable(gwas_hits_table, caption = "Case mQTL SNPs within 250kb of a known Nalls et al. PD GWAS locus")
```

Case mQTL SNPs within 250kb of a known Nalls et al. PD GWAS locus

snp_id	gwas_nearest_gene	gwas_locus_number	gwas_dist_bp
6:112243291	FYN	33	0
6:30108751	TRIM40	30	68
4:77148213	FAM47E	22	244
6:30108336	TRIM40	30	347
6:30109063	TRIM40	30	380
2:169109956	STK39	11	438
4:77148601	FAM47E	22	632
4:77148761	FAM47E	22	792
6:112242253	FYN	33	1038
4:77149119	FAM47E	22	1150
2:169109230	STK39	11	1164
6:30107309	TRIM40	30	1374
6:30107153	TRIM40	30	1530
4:77149540	FAM47E	22	1571
4:77149772	FAM47E	22	1803
4:77145892	FAM47E	22	2077
6:30106493	TRIM40	30	2190
6:30106202	TRIM40	30	2481
4:77150468	FAM47E	22	2499
13:49930369	CAB39L	53	2637
4:77150655	FAM47E	22	2686
2:169107532	STK39	11	2862
4:77150838	FAM47E	22	2869
4:77151109	FAM47E	22	3140
4:77151110	FAM47E	22	3141
6:30105391	TRIM40	30	3292
4:77151300	FAM47E	22	3331
2:169107007	STK39	11	3387
2:169113808	STK39	11	3414
2:169106850	STK39	11	3544
2:169114387	STK39	11	3993
2:169115051	STK39	11	4657
4:957284	TMEM175	19	5337
2:169116027	STK39	11	5633
6:30114402	TRIM40	30	5719
2:169116312	STK39	11	5918

snp_id	gwas_nearest_gene	gwas_locus_number	gwas_dist_bp
2:169117008	STK39	11	6614
6:112250110	FYN	33	6819
4:77154957	FAM47E	22	6988
4:77155139	FAM47E	22	7170
2:169103143	STK39	11	7251
2:169117706	STK39	11	7312
6:30116078	TRIM40	30	7395
2:169117820	STK39	11	7426
4:77155486	FAM47E	22	7517
2:169118135	STK39	11	7741
2:169102577	STK39	11	7817
2:169118415	STK39	11	8021
2:169118454	STK39	11	8060
6:32570401	HLA-DRB5	31	8371
2:169119257	STK39	11	8863
2:169119317	STK39	11	8923
2:169120704	STK39	11	10310
2:169121700	STK39	11	11306
2:169121841	STK39	11	11447
6:30096772	TRIM40	30	11911
2:169122597	STK39	11	12203
6:30120912	TRIM40	30	12229
2:169122644	STK39	11	12250
2:169122942	STK39	11	12548
6:32591609	HLA-DRB5	31	12837
4:77161299	FAM47E	22	13330
6:30095202	TRIM40	30	13481
6:30094315	TRIM40	30	14368
6:30094113	TRIM40	30	14570
2:169125080	STK39	11	14686
2:169125462	STK39	11	15068
13:49943607	CAB39L	53	15875
2:169126764	STK39	11	16370
1:155151493	KRTCAP2	1	16457
4:77131294	FAM47E	22	16675
1:155151754	KRTCAP2	1	16718
6:30125543	TRIM40	30	16860
4:77165605	FAM47E	22	17636
6:30127805	TRIM40	30	19122
2:169129601	STK39	11	19207
2:169129878	STK39	11	19484
2:169090729	STK39	11	19665
4:77168112	FAM47E	22	20143

snp_id	gwas_nearest_gene	gwas_locus_number	gwas_dist_bp
6:30128949	TRIM40	30	20266
4:77168321	FAM47E	22	20352
2:169089920	STK39	11	20474
1:155155731	KRTCAP2	1	20695
6:32599485	HLA-DRB5	31	20713
6:32599999	HLA-DRB5	31	21227
2:169131721	STK39	11	21327
6:32600317	HLA-DRB5	31	21545
6:32600404	HLA-DRB5	31	21632
6:32600448	HLA-DRB5	31	21676
6:112221611	FYN	33	21680
6:32600914	HLA-DRB5	31	22142
19:2363319	SPPL2B	76	22272
1:155157635	KRTCAP2	1	22599
2:169133002	STK39	11	22608
1:155157715	KRTCAP2	1	22679
2:169133371	STK39	11	22977
6:112266347	FYN	33	23056
6:32602238	HLA-DRB5	31	23466
6:112267233	FYN	33	23942
4:77171925	FAM47E	22	23956
6:30084696	TRIM40	30	23987
6:30084549	TRIM40	30	24134
6:30084505	TRIM40	30	24178
6:30084446	TRIM40	30	24237
4:77173739	FAM47E-STBD1	22	24315
6:32603098	HLA-DRB5	31	24326
4:77172445	FAM47E	22	24476
6:112218538	FYN	33	24753
6:112218167	FYN	33	25124
13:49901088	CAB39L	53	26644
6:112216413	FYN	33	26878
6:112215441	FYN	33	27850
6:112272725	FYN	33	29434
13:49897778	CAB39L	53	29954
6:112213313	FYN	33	29978
6:112212934	FYN	33	30357
13:49897226	CAB39L	53	30506
2:169140909	STK39	11	30515
14:88433660	GALC	58	30604
6:112212449	FYN	33	30842
6:112211802	FYN	33	31489
6:30140342	TRIM40	30	31659

snp_id	gwas_nearest_gene	gwas_locus_number	gwas_dist_bp
13:49895644	CAB39L	53	32088
6:30076426	TRIM40	30	32257
6:30076163	TRIM40	30	32520
6:30076064	TRIM40	30	32619
6:30076018	TRIM40	30	32665
6:30076016	TRIM40	30	32667
6:30075843	TRIM40	30	32840
6:112210443	FYN	33	32848
13:49894824	CAB39L	53	32908
6:30075641	TRIM40	30	33042
13:49894274	CAB39L	53	33458
6:30074873	TRIM40	30	33810
1:155168930	KRTCAP2	1	33894
1:155169355	KRTCAP2	1	34319
6:112208753	FYN	33	34538
13:49893105	CAB39L	53	34627
13:49893024	CAB39L	53	34708
6:30073847	TRIM40	30	34836
6:30073776	TRIM40	30	34907
6:112208191	FYN	33	35100
12:133028325	FBRSL1	52	35443
13:49891933	CAB39L	53	35799
13:49891231	CAB39L	53	36501
13:49890902	CAB39L	53	36830
6:30146019	TRIM40	30	37336
13:49889267	CAB39L	53	38465
2:169071190	STK39	11	39204
21:38891587	DYRK1A	78	39226
6:112203750	FYN	33	39541
14:88424120	GALC	58	40144
6:112203035	FYN	33	40256
6:30068195	TRIM40	30	40488
2:169069717	STK39	11	40677
13:49886855	CAB39L	53	40877
13:49886656	CAB39L	53	41076
2:169068506	STK39	11	41888
6:112200061	FYN	33	43230
6:112200021	FYN	33	43270
13:49884055	CAB39L	53	43677
6:30064702	TRIM40	30	43981
13:49883444	CAB39L	53	44288
17:7310473	CHRNB1	66	45148
2:169065204	STK39	11	45190

snp_id	gwas_nearest_gene	gwas_locus_number	gwas_dist_bp
2:169156742	STK39	11	46348
14:88417821	GALC	58	46443
6:32626537	HLA-DRB5	31	47765
6:112194496	FYN	33	48795
13:49878909	CAB39L	53	48823
13:49878804	CAB39L	53	48928
6:32627700	HLA-DRB5	31	48928
13:49878714	CAB39L	53	49018
13:49878236	CAB39L	53	49496
11:133737462	IGSF9B	48	49539
11:133737304	IGSF9B	48	49697
2:169060668	STK39	11	49726
13:49877951	CAB39L	53	49781
5:60088148	ELOVL7	26	49811
13:49877403	CAB39L	53	50329
2:169160724	STK39	11	50330
6:112192795	FYN	33	50496
13:49876659	CAB39L	53	51073
11:133735810	IGSF9B	48	51191
13:49876171	CAB39L	53	51561
11:133734322	IGSF9B	48	52679
11:133734177	IGSF9B	48	52824
2:169163422	STK39	11	53028
11:133733893	IGSF9B	48	53108
11:133733864	IGSF9B	48	53137
6:112189712	FYN	33	53579
13:49873894	CAB39L	53	53838
6:112189294	FYN	33	53997
13:49873678	CAB39L	53	54054
11:133732265	IGSF9B	48	54736
11:133732150	IGSF9B	48	54851
17:7300611	CHRNB1	66	55010
2:169165476	STK39	11	55082
11:133731644	IGSF9B	48	55357
17:44922276	WNT3	70	55471
6:112187553	FYN	33	55738
11:133731134	IGSF9B	48	55867
11:133730820	IGSF9B	48	56181
6:112185838	FYN	33	57453
11:133729469	IGSF9B	48	57532
6:112185732	FYN	33	57559
13:49869578	CAB39L	53	58154
2:18089614	KCNS3	7	58234

snp_id	gwas_nearest_gene	gwas_locus_number	gwas_dist_bp
13:49869445	CAB39L	53	58287
6:30167294	TRIM40	30	58611
6:112184621	FYN	33	58670
6:112184520	FYN	33	58771
2:169051157	STK39	11	59237
13:49868341	CAB39L	53	59391
11:133727535	IGSF9B	48	59466
17:44926818	WNT3	70	60013
13:49867290	CAB39L	53	60442
17:44927282	WNT3	70	60477
17:44927374	WNT3	70	60569
11:133726067	IGSF9B	48	60934
17:44927816	WNT3	70	61011
11:133725989	IGSF9B	48	61012
6:112182111	FYN	33	61180
17:44928053	WNT3	70	61248
13:49866322	CAB39L	53	61410
17:44928769	WNT3	70	61964
6:30046682	TRIM40	30	62001
6:30046429	TRIM40	30	62254
6:30046376	TRIM40	30	62307
6:30046011	TRIM40	30	62672
6:30046005	TRIM40	30	62678
13:49864372	CAB39L	53	63360
17:44930345	WNT3	70	63540
17:44930363	WNT3	70	63558
13:49864107	CAB39L	53	63625
13:49864106	CAB39L	53	63626
6:30044827	TRIM40	30	63856
13:49863811	CAB39L	53	63921
6:112179226	FYN	33	64065
2:18083460	KCNS3	7	64388
6:30044249	TRIM40	30	64434
17:44931434	WNT3	70	64629
6:30043992	TRIM40	30	64691
6:30043955	TRIM40	30	64728
6:30043779	TRIM40	30	64904
13:49862027	CAB39L	53	65705
6:30042861	TRIM40	30	65822
13:49861155	CAB39L	53	66577
6:30042057	TRIM40	30	66626
11:133720077	IGSF9B	48	66924
13:49860691	CAB39L	53	67041

snp_id	gwas_nearest_gene	gwas_locus_number	gwas_dist_bp
6:112176239	FYN	33	67052
6:30041509	TRIM40	30	67174
2:169043083	STK39	11	67311
11:133719565	IGSF9B	48	67436
6:112175735	FYN	33	67556
6:112175517	FYN	33	67774
11:133719221	IGSF9B	48	67780
11:133718740	IGSF9B	48	68261
11:133718467	IGSF9B	48	68534
6:30039767	TRIM40	30	68916
13:49858731	CAB39L	53	69001
11:133717902	IGSF9B	48	69099
1:205653756	NUCKS1	4	69816
6:30038823	TRIM40	30	69860
6:112172975	FYN	33	70316
6:30179164	TRIM40	30	70481
6:112172746	FYN	33	70545
11:133716356	IGSF9B	48	70645
6:30037908	TRIM40	30	70775
2:18077007	KCNS3	7	70841
6:30037686	TRIM40	30	70997
11:133715791	IGSF9B	48	71210
13:49856198	CAB39L	53	71534
11:133715153	IGSF9B	48	71848
6:112171130	FYN	33	72161
11:133714672	IGSF9B	48	72329
13:49855276	CAB39L	53	72456
11:133714522	IGSF9B	48	72479
13:49854515	CAB39L	53	73217
11:133713730	IGSF9B	48	73271
11:133713436	IGSF9B	48	73565
2:169036789	STK39	11	73605
11:133713189	IGSF9B	48	73812
11:133713094	IGSF9B	48	73907
11:133713082	IGSF9B	48	73919
6:112169138	FYN	33	74153
2:169036091	STK39	11	74303
6:30034289	TRIM40	30	74394
2:169035989	STK39	11	74405
11:133712594	IGSF9B	48	74407
11:133712473	IGSF9B	48	74528
11:133712348	IGSF9B	48	74653
2:169035673	STK39	11	74721

snp_id	gwas_nearest_gene	gwas_locus_number	gwas_dist_bp
6:112168309	FYN	33	74982
13:49852430	CAB39L	53	75302
13:49852377	CAB39L	53	75355
13:49852000	CAB39L	53	75732
6:30032673	TRIM40	30	76010
13:49850894	CAB39L	53	76838
13:49850649	CAB39L	53	77083
6:112165576	FYN	33	77715
6:112164441	FYN	33	78850
13:49848833	CAB39L	53	78899
13:49848794	CAB39L	53	78938
6:112164313	FYN	33	78978
13:49848194	CAB39L	53	79538
13:49847452	CAB39L	53	80280
13:49846920	CAB39L	53	80812
13:49846230	CAB39L	53	81502
6:112161786	FYN	33	81505
6:112161092	FYN	33	82199
13:49845485	CAB39L	53	82247
6:30026236	TRIM40	30	82447
2:169027555	STK39	11	82839
13:49844085	CAB39L	53	83647
13:49843964	CAB39L	53	83768
17:40657101	RETREG3	67	83912
6:112158134	FYN	33	85157
6:112157615	FYN	33	85676
6:112157097	FYN	33	86194
13:49841499	CAB39L	53	86233
6:112156570	FYN	33	86721
6:112156463	FYN	33	86828
17:40653534	RETREG3	67	87479
6:112155789	FYN	33	87502
6:112155722	FYN	33	87569
13:49839979	CAB39L	53	87753
6:112155473	FYN	33	87818
13:49839889	CAB39L	53	87843
13:49839548	CAB39L	53	88184
13:49838972	CAB39L	53	88760
6:112154244	FYN	33	89047
12:133152840	FBRSL1	52	89072
6:112154200	FYN	33	89091
6:112154021	FYN	33	89270
2:169020642	STK39	11	89752

snp_id	gwas_nearest_gene	gwas_locus_number	gwas_dist_bp
6:112153422	FYN	33	89869
17:40650805	RETREG3	67	90208
6:30199016	TRIM40	30	90333
6:112152768	FYN	33	90523
13:49837022	CAB39L	53	90710
6:112152560	FYN	33	90731
6:112152494	FYN	33	90797
6:112152415	FYN	33	90876
19:2432177	SPPL2B	76	91130
19:2432781	SPPL2B	76	91734
6:112151452	FYN	33	91839
6:112151444	FYN	33	91847
6:30016297	TRIM40	30	92386
12:133156187	FBRSL1	52	92419
6:112150430	FYN	33	92861
13:49834463	CAB39L	53	93269
13:49834252	CAB39L	53	93480
6:112149486	FYN	33	93805
4:1045824	TMEM175	19	93877
6:112149373	FYN	33	93918
13:49833774	CAB39L	53	93958
2:169016263	STK39	11	94131
2:169016171	STK39	11	94223
19:2435334	SPPL2B	76	94287
6:112148293	FYN	33	94998
4:1046992	TMEM175	19	95045
19:2436424	SPPL2B	76	95377
6:112147909	FYN	33	95382
19:2437058	SPPL2B	76	96011
13:49831703	CAB39L	53	96029
6:112147107	FYN	33	96184
13:49830629	CAB39L	53	97103
2:169012955	STK39	11	97439
19:2439163	SPPL2B	76	98116
19:2439186	SPPL2B	76	98139
19:2439215	SPPL2B	76	98168
19:2439319	SPPL2B	76	98272
6:112143472	FYN	33	99819
4:1052009	TMEM175	19	100062
13:49826508	CAB39L	53	101224
6:112141908	FYN	33	101383
6:112141732	FYN	33	101559
1:154796342	PMVK	1	101843

snp_id	gwas_nearest_gene	gwas_locus_number	gwas_dist_bp
13:49825667	CAB39L	53	102065
17:76323344	DNAH17	72	102136
13:49825360	CAB39L	53	102372
13:49824356	CAB39L	53	103376
6:112138882	FYN	33	104409
6:112138294	FYN	33	104997
6:32685865	HLA-DRB5	31	107093
2:169001183	STK39	11	109211
17:7245674	CHRNB1	66	109947
2:169000293	STK39	11	110101
6:112132032	FYN	33	111259
6:112131648	FYN	33	111643
6:112131578	FYN	33	111713
6:112130478	FYN	33	112813
6:112129636	FYN	33	113655
5:60024091	ELOVL7	26	113868
6:112128947	FYN	33	114344
12:133178417	FBRSL1	52	114649
8:11828200	CTSB	37	115757
6:30224970	TRIM40	30	116287
6:112126303	FYN	33	116988
12:133181399	FBRSL1	52	117631
12:133181425	FBRSL1	52	117657
8:11830150	CTSB	37	117707
6:112125453	FYN	33	117838
6:112125335	FYN	33	117956
6:112124440	FYN	33	118851
6:112124438	FYN	33	118853
6:32698391	HLA-DRB5	31	119619
6:72367279	RIMS1	32	120483
6:112122226	FYN	33	121065
6:112121766	FYN	33	121525
6:112121228	FYN	33	122063
6:112121096	FYN	33	122195
17:7478402	CHRNB1	66	122781
6:112119981	FYN	33	123310
6:112119619	FYN	33	123672
6:32703407	HLA-DRB5	31	124635
6:112118505	FYN	33	124786
6:29983298	TRIM40	30	125385
5:60012521	ELOVL7	26	125438
6:112116451	FYN	33	126840
6:112115993	FYN	33	127298

snp_id	gwas_nearest_gene	gwas_locus_number	gwas_dist_bp
6:112115355	FYN	33	127936
6:32706801	HLA-DRB5	31	128029
17:44994859	WNT3	70	128054
6:112114791	FYN	33	128500
6:112114308	FYN	33	128983
6:112114249	FYN	33	129042
6:112114151	FYN	33	129140
6:112113958	FYN	33	129333
6:112113938	FYN	33	129353
6:29977869	TRIM40	30	130814
6:112111394	FYN	33	131897
6:112110528	FYN	33	132763
4:790757	GAK	19	134619
17:43933307	CRHR1	69	134999
4:790182	GAK	19	135194
6:32442086	HLA-DRB5	31	136686
17:7492388	CHRNB1	66	136767
6:32441555	HLA-DRB5	31	137217
6:32441408	HLA-DRB5	31	137364
6:32440910	HLA-DRB5	31	137862
6:32440879	HLA-DRB5	31	137893
6:32440467	HLA-DRB5	31	138305
6:32440451	HLA-DRB5	31	138321
6:32440362	HLA-DRB5	31	138410
6:32439964	HLA-DRB5	31	138808
6:32718345	HLA-DRB5	31	139573
6:30248711	TRIM40	30	140028
6:29966726	TRIM40	30	141957
6:30251391	TRIM40	30	142708
6:32435044	HLA-DRB5	31	143728
6:32724043	HLA-DRB5	31	145271
6:32432509	HLA-DRB5	31	146263
6:32431867	HLA-DRB5	31	146905
6:32431606	HLA-DRB5	31	147166
6:32431147	HLA-DRB5	31	147625
6:29960420	TRIM40	30	148263
6:32430289	HLA-DRB5	31	148483
6:32429594	HLA-DRB5	31	149178
6:32427748	HLA-DRB5	31	151024
15:61845512	VPS13C	59	151873
3:182912282	MCCC1	18	152209
3:182912526	MCCC1	18	152453
3:182913092	MCCC1	18	153019

snp_id	gwas_nearest_gene	gwas_locus_number	gwas_dist_bp
6:32424594	HLA-DRB5	31	154178
6:32734304	HLA-DRB5	31	155532
6:32736079	HLA-DRB5	31	157307
6:32740671	HLA-DRB5	31	161899
6:32741006	HLA-DRB5	31	162234
6:32741742	HLA-DRB5	31	162970
6:32741894	HLA-DRB5	31	163122
6:32742059	HLA-DRB5	31	163287
6:32742531	HLA-DRB5	31	163759
6:32414435	HLA-DRB5	31	164337
6:32743173	HLA-DRB5	31	164401
6:32743193	HLA-DRB5	31	164421
6:32743217	HLA-DRB5	31	164445
6:32743861	HLA-DRB5	31	165089
6:32744050	HLA-DRB5	31	165278
6:32744145	HLA-DRB5	31	165373
6:32744321	HLA-DRB5	31	165549
6:32744452	HLA-DRB5	31	165680
6:32744566	HLA-DRB5	31	165794
6:32744667	HLA-DRB5	31	165895
6:32744684	HLA-DRB5	31	165912
6:32744730	HLA-DRB5	31	165958
6:29942527	TRIM40	30	166156
6:32745156	HLA-DRB5	31	166384
6:32745324	HLA-DRB5	31	166552
6:32746012	HLA-DRB5	31	167240
6:32746845	HLA-DRB5	31	168073
6:32746920	HLA-DRB5	31	168148
6:32746984	HLA-DRB5	31	168212
6:32747012	HLA-DRB5	31	168240
6:32747060	HLA-DRB5	31	168288
6:32747273	HLA-DRB5	31	168501
6:29939677	TRIM40	30	169006
6:32747959	HLA-DRB5	31	169187
6:32747971	HLA-DRB5	31	169199
6:32748633	HLA-DRB5	31	169861
6:32408497	HLA-DRB5	31	170275
6:30279130	TRIM40	30	170447
6:32749704	HLA-DRB5	31	170932
6:29937555	TRIM40	30	171128
6:32750031	HLA-DRB5	31	171259
6:32750279	HLA-DRB5	31	171507
6:32750982	HLA-DRB5	31	172210

snp_id	gwas_nearest_gene	gwas_locus_number	gwas_dist_bp
6:32752566	HLA-DRB5	31	173794
13:50101526	CAB39L	53	173794
6:32752577	HLA-DRB5	31	173805
6:32752788	HLA-DRB5	31	174016
6:32753313	HLA-DRB5	31	174541
6:29934109	TRIM40	30	174574
6:32753492	HLA-DRB5	31	174720
6:32753682	HLA-DRB5	31	174910
6:32753739	HLA-DRB5	31	174967
6:133035325	RPS12	34	175036
7:23475364	GPNMB	35	175315
6:32754094	HLA-DRB5	31	175322
6:29932897	TRIM40	30	175786
6:32754920	HLA-DRB5	31	176148
6:32755023	HLA-DRB5	31	176251
6:32755216	HLA-DRB5	31	176444
6:30285312	TRIM40	30	176629
6:32755937	HLA-DRB5	31	177165
6:32756363	HLA-DRB5	31	177591
6:133032759	RPS12	34	177602
6:32756872	HLA-DRB5	31	178100
6:32756874	HLA-DRB5	31	178102
6:32756961	HLA-DRB5	31	178189
6:32757131	HLA-DRB5	31	178359
6:32757158	HLA-DRB5	31	178386
6:32757396	HLA-DRB5	31	178624
6:32757453	HLA-DRB5	31	178681
6:133031252	RPS12	34	179109
6:32757938	HLA-DRB5	31	179166
6:133031111	RPS12	34	179250
6:133030711	RPS12	34	179650
6:133030710	RPS12	34	179651
6:133030492	RPS12	34	179869
6:32758966	HLA-DRB5	31	180194
6:32760249	HLA-DRB5	31	181477
6:32760257	HLA-DRB5	31	181485
6:32760334	HLA-DRB5	31	181562
6:32760539	HLA-DRB5	31	181767
6:32760655	HLA-DRB5	31	181883
7:23482044	GPNMB	35	181995
6:32760965	HLA-DRB5	31	182193
4:743125	GAK	19	182251
6:32761317	HLA-DRB5	31	182545

snp_id	gwas_nearest_gene	gwas_locus_number	gwas_dist_bp
6:32761507	HLA-DRB5	31	182735
6:32761776	HLA-DRB5	31	183004
6:32395726	HLA-DRB5	31	183046
6:32761826	HLA-DRB5	31	183054
7:23483373	GPNMB	35	183324
6:32762103	HLA-DRB5	31	183331
6:32762226	HLA-DRB5	31	183454
6:32762307	HLA-DRB5	31	183535
6:32762484	HLA-DRB5	31	183712
6:32762616	HLA-DRB5	31	183844
6:32762654	HLA-DRB5	31	183882
6:32763040	HLA-DRB5	31	184268
6:133025684	RPS12	34	184677
6:133025637	RPS12	34	184724
6:32763514	HLA-DRB5	31	184742
6:32763521	HLA-DRB5	31	184749
6:32763821	HLA-DRB5	31	185049
6:32763974	HLA-DRB5	31	185202
6:29923351	TRIM40	30	185332
6:133024980	RPS12	34	185381
6:29922885	TRIM40	30	185798
7:23485919	GPNMB	35	185870
6:32764719	HLA-DRB5	31	185947
6:133024183	RPS12	34	186178
6:32765383	HLA-DRB5	31	186611
6:29921999	TRIM40	30	186684
6:29921460	TRIM40	30	187223
6:72300534	RIMS1	32	187228
6:133023036	RPS12	34	187325
6:133022761	RPS12	34	187600
6:133022468	RPS12	34	187893
6:133022197	RPS12	34	188164
6:32767091	HLA-DRB5	31	188319
6:29919819	TRIM40	30	188864
7:23490230	GPNMB	35	190181
6:32768980	HLA-DRB5	31	190208
6:30299245	TRIM40	30	190562
6:32387432	HLA-DRB5	31	191340
6:32770390	HLA-DRB5	31	191618
6:29916971	TRIM40	30	191712
6:29916922	TRIM40	30	191761
6:29916654	TRIM40	30	192029
6:29916639	TRIM40	30	192044

snp_id	gwas_nearest_gene	gwas_locus_number	gwas_dist_bp
6:29916480	TRIM40	30	192203
6:32771354	HLA-DRB5	31	192582
6:32771531	HLA-DRB5	31	192759
6:32771610	HLA-DRB5	31	192838
6:29915839	TRIM40	30	192844
6:29915791	TRIM40	30	192892
6:29915694	TRIM40	30	192989
6:32771887	HLA-DRB5	31	193115
6:32772004	HLA-DRB5	31	193232
6:29915438	TRIM40	30	193245
17:42627902	FAM171A2	68	193272
6:133017042	RPS12	34	193319
6:29915351	TRIM40	30	193332
6:133016967	RPS12	34	193394
6:133016902	RPS12	34	193459
6:133016685	RPS12	34	193676
6:32772636	HLA-DRB5	31	193864
6:133016337	RPS12	34	194024
6:133016300	RPS12	34	194061
6:29914553	TRIM40	30	194130
6:29914544	TRIM40	30	194139
6:29914478	TRIM40	30	194205
6:133016151	RPS12	34	194210
6:32773042	HLA-DRB5	31	194270
6:133016000	RPS12	34	194361
6:32773172	HLA-DRB5	31	194400
6:133015951	RPS12	34	194410
6:133015756	RPS12	34	194605
6:29914064	TRIM40	30	194619
6:133015662	RPS12	34	194699
6:29913886	TRIM40	30	194797
6:29913830	TRIM40	30	194853
6:29913824	TRIM40	30	194859
6:29913698	TRIM40	30	194985
17:42629737	FAM171A2	68	195107
6:29913449	TRIM40	30	195234
6:32774385	HLA-DRB5	31	195613
4:15541609	BST1	20	195739
6:29912938	TRIM40	30	195745
6:29912713	TRIM40	30	195970
6:32774788	HLA-DRB5	31	196016
6:29912386	TRIM40	30	196297
6:29912297	TRIM40	30	196386

snp_id	gwas_nearest_gene	gwas_locus_number	gwas_dist_bp
6:29912281	TRIM40	30	196402
6:32775465	HLA-DRB5	31	196693
6:32775563	HLA-DRB5	31	196791
6:32775602	HLA-DRB5	31	196830
6:32775686	HLA-DRB5	31	196914
6:32775821	HLA-DRB5	31	197049
6:29911604	TRIM40	30	197079
6:29911571	TRIM40	30	197112
6:29911545	TRIM40	30	197138
6:32776120	HLA-DRB5	31	197348
6:32381280	HLA-DRB5	31	197492
6:32776414	HLA-DRB5	31	197642
6:29910947	TRIM40	30	197736
6:32776623	HLA-DRB5	31	197851
6:32776832	HLA-DRB5	31	198060
6:32776854	HLA-DRB5	31	198082
6:32776976	HLA-DRB5	31	198204
6:29910402	TRIM40	30	198281
6:32777140	HLA-DRB5	31	198368
6:32777198	HLA-DRB5	31	198426
6:32777216	HLA-DRB5	31	198444
6:32777305	HLA-DRB5	31	198533
6:29909999	TRIM40	30	198684
6:29909988	TRIM40	30	198695
6:32777472	HLA-DRB5	31	198700
6:32777533	HLA-DRB5	31	198761
6:32778052	HLA-DRB5	31	199280
6:32778153	HLA-DRB5	31	199381
6:32778531	HLA-DRB5	31	199759
6:29908892	TRIM40	30	199791
6:29908110	TRIM40	30	200573
6:32780201	HLA-DRB5	31	201429
6:32780317	HLA-DRB5	31	201545
6:29906489	TRIM40	30	202194
6:32375695	HLA-DRB5	31	203077
6:32373829	HLA-DRB5	31	204943
6:32373185	HLA-DRB5	31	205587
6:32372963	HLA-DRB5	31	205809
6:32372863	HLA-DRB5	31	205909
6:32372635	HLA-DRB5	31	206137
6:32372236	HLA-DRB5	31	206536
6:32785515	HLA-DRB5	31	206743
6:32369886	HLA-DRB5	31	208886

snp_id	gwas_nearest_gene	gwas_locus_number	gwas_dist_bp
6:32369643	HLA-DRB5	31	209129
6:32369355	HLA-DRB5	31	209417
6:30318327	TRIM40	30	209644
6:32368462	HLA-DRB5	31	210310
6:30319154	TRIM40	30	210471
6:32368087	HLA-DRB5	31	210685
6:32367927	HLA-DRB5	31	210845
6:32367847	HLA-DRB5	31	210925
6:32367604	HLA-DRB5	31	211168
6:32367515	HLA-DRB5	31	211257
6:32367451	HLA-DRB5	31	211321
6:32367259	HLA-DRB5	31	211513
6:32366625	HLA-DRB5	31	212147
6:32790940	HLA-DRB5	31	212168
6:32366421	HLA-DRB5	31	212351
6:32366331	HLA-DRB5	31	212441
6:32365971	HLA-DRB5	31	212801
6:32365896	HLA-DRB5	31	212876
6:32365707	HLA-DRB5	31	213065
4:712222	GAK	19	213154
4:712048	GAK	19	213328
17:40527544	RETREG3	67	213469
6:32365284	HLA-DRB5	31	213488
6:32364984	HLA-DRB5	31	213788
6:30323393	TRIM40	30	214710
17:40526273	RETREG3	67	214740
7:23514868	GPNMB	35	214819
6:32793770	HLA-DRB5	31	214998
6:32363774	HLA-DRB5	31	214998
1:161684890	FCGR2A	2	215836
6:32362930	HLA-DRB5	31	215842
17:40525098	RETREG3	67	215915
6:32362832	HLA-DRB5	31	215940
6:32362745	HLA-DRB5	31	216027
6:32362741	HLA-DRB5	31	216031
6:32362669	HLA-DRB5	31	216103
6:32362639	HLA-DRB5	31	216133
6:32362453	HLA-DRB5	31	216319
6:32362389	HLA-DRB5	31	216383
6:32362272	HLA-DRB5	31	216500
6:32362262	HLA-DRB5	31	216510
6:32362217	HLA-DRB5	31	216555
6:32362182	HLA-DRB5	31	216590

snp_id	gwas_nearest_gene	gwas_locus_number	gwas_dist_bp
6:32362128	HLA-DRB5	31	216644
6:32362085	HLA-DRB5	31	216687
6:32362075	HLA-DRB5	31	216697
6:32362020	HLA-DRB5	31	216752
6:32362010	HLA-DRB5	31	216762
6:32361953	HLA-DRB5	31	216819
6:32361943	HLA-DRB5	31	216829
6:32361894	HLA-DRB5	31	216878
6:32361762	HLA-DRB5	31	217010
6:32361694	HLA-DRB5	31	217078
6:32361583	HLA-DRB5	31	217189
6:32361566	HLA-DRB5	31	217206
6:32361525	HLA-DRB5	31	217247
6:32361485	HLA-DRB5	31	217287
6:32361424	HLA-DRB5	31	217348
6:32361388	HLA-DRB5	31	217384
7:23517468	GPNMB	35	217419
6:32361255	HLA-DRB5	31	217517
6:32361251	HLA-DRB5	31	217521
6:32361223	HLA-DRB5	31	217549
6:32361130	HLA-DRB5	31	217642
6:32361080	HLA-DRB5	31	217692
6:32361035	HLA-DRB5	31	217737
6:32361003	HLA-DRB5	31	217769
6:32360970	HLA-DRB5	31	217802
6:32360955	HLA-DRB5	31	217817
6:32360945	HLA-DRB5	31	217827
6:32360714	HLA-DRB5	31	218058
6:30326804	TRIM40	30	218121
6:30326809	TRIM40	30	218126
6:32360644	HLA-DRB5	31	218128
6:32360641	HLA-DRB5	31	218131
6:32360589	HLA-DRB5	31	218183
6:32360551	HLA-DRB5	31	218221
6:32360532	HLA-DRB5	31	218240
6:32360477	HLA-DRB5	31	218295
6:32360443	HLA-DRB5	31	218329
6:32360341	HLA-DRB5	31	218431
6:30327194	TRIM40	30	218511
6:30327196	TRIM40	30	218513
12:133282304	FBRSL1	52	218536
6:32360136	HLA-DRB5	31	218636
6:32360135	HLA-DRB5	31	218637

snp_id	gwas_nearest_gene	gwas_locus_number	gwas_dist_bp
6:32360069	HLA-DRB5	31	218703
6:32360060	HLA-DRB5	31	218712
6:32360050	HLA-DRB5	31	218722
6:32359904	HLA-DRB5	31	218868
6:32359800	HLA-DRB5	31	218972
6:32359617	HLA-DRB5	31	219155
6:30327839	TRIM40	30	219156
7:23519260	GPNMB	35	219211
6:32359460	HLA-DRB5	31	219312
6:32359431	HLA-DRB5	31	219341
6:32359389	HLA-DRB5	31	219383
6:30328192	TRIM40	30	219509
7:23519647	GPNMB	35	219598
6:32359133	HLA-DRB5	31	219639
7:23519710	GPNMB	35	219661
6:30328357	TRIM40	30	219674
6:32798548	HLA-DRB5	31	219776
6:32358883	HLA-DRB5	31	219889
6:32358834	HLA-DRB5	31	219938
6:32358728	HLA-DRB5	31	220044
7:23520104	GPNMB	35	220055
6:32358533	HLA-DRB5	31	220239
6:32358459	HLA-DRB5	31	220313
6:32358368	HLA-DRB5	31	220404
6:32358345	HLA-DRB5	31	220427
6:32358213	HLA-DRB5	31	220559
6:32358205	HLA-DRB5	31	220567
6:32358201	HLA-DRB5	31	220571
6:32358163	HLA-DRB5	31	220609
6:32358144	HLA-DRB5	31	220628
6:32358113	HLA-DRB5	31	220659
6:32358011	HLA-DRB5	31	220761
6:32358008	HLA-DRB5	31	220764
4:704602	GAK	19	220774
7:23520963	GPNMB	35	220914
6:32357715	HLA-DRB5	31	221057
6:30329926	TRIM40	30	221243
6:32357529	HLA-DRB5	31	221243
6:32357525	HLA-DRB5	31	221247
7:23521316	GPNMB	35	221267
6:30329966	TRIM40	30	221283
6:32357489	HLA-DRB5	31	221283
6:32357397	HLA-DRB5	31	221375

snp_id	gwas_nearest_gene	gwas_locus_number	gwas_dist_bp
6:30330071	TRIM40	30	221388
6:32357344	HLA-DRB5	31	221428
6:32357339	HLA-DRB5	31	221433
6:32357333	HLA-DRB5	31	221439
6:32357133	HLA-DRB5	31	221639
6:30330392	TRIM40	30	221709
6:32357037	HLA-DRB5	31	221735
6:32357023	HLA-DRB5	31	221749
6:32356917	HLA-DRB5	31	221855
6:32356876	HLA-DRB5	31	221896
6:32356839	HLA-DRB5	31	221933
6:32356829	HLA-DRB5	31	221943
6:32356755	HLA-DRB5	31	222017
6:30330737	TRIM40	30	222054
6:32356667	HLA-DRB5	31	222105
6:32356573	HLA-DRB5	31	222199
6:32356489	HLA-DRB5	31	222283
6:32356440	HLA-DRB5	31	222332
6:32356356	HLA-DRB5	31	222416
6:32356307	HLA-DRB5	31	222465
6:32356272	HLA-DRB5	31	222500
6:32356242	HLA-DRB5	31	222530
6:30331216	TRIM40	30	222533
6:32356097	HLA-DRB5	31	222675
6:32355981	HLA-DRB5	31	222791
6:32355858	HLA-DRB5	31	222914
6:32355724	HLA-DRB5	31	223048
6:32355677	HLA-DRB5	31	223095
6:32355631	HLA-DRB5	31	223141
6:32355303	HLA-DRB5	31	223469
6:32355285	HLA-DRB5	31	223487
6:32355109	HLA-DRB5	31	223663
6:32355003	HLA-DRB5	31	223769
6:32354888	HLA-DRB5	31	223884
6:32354834	HLA-DRB5	31	223938
6:32354629	HLA-DRB5	31	224143
6:32354520	HLA-DRB5	31	224252
6:30333505	TRIM40	30	224822
6:32352294	HLA-DRB5	31	226478
6:30335350	TRIM40	30	226667
17:40514201	RETREG3	67	226812
6:30336663	TRIM40	30	227980
6:32350591	HLA-DRB5	31	228181

snp_id	gwas_nearest_gene	gwas_locus_number	gwas_dist_bp
6:32350384	HLA-DRB5	31	228388
6:30337141	TRIM40	30	228458
6:32349557	HLA-DRB5	31	229215
6:30337974	TRIM40	30	229291
8:22294836	BIN3	39	231144
6:30340528	TRIM40	30	231845
6:30341390	TRIM40	30	232707
6:32345689	HLA-DRB5	31	233083
6:32345131	HLA-DRB5	31	233641
8:22291265	BIN3	39	234715
6:32343743	HLA-DRB5	31	235029
6:32343369	HLA-DRB5	31	235403
6:32343339	HLA-DRB5	31	235433
6:32343236	HLA-DRB5	31	235536
8:22289582	BIN3	39	236398
8:22289288	BIN3	39	236692
8:22289166	BIN3	39	236814
8:22289126	BIN3	39	236854
8:22289093	BIN3	39	236887
8:22288995	BIN3	39	236985
8:22288889	BIN3	39	237091
8:22288815	BIN3	39	237165
8:22288789	BIN3	39	237191
8:22288778	BIN3	39	237202
8:22288637	BIN3	39	237343
8:22288342	BIN3	39	237638
8:22288259	BIN3	39	237721
8:22288242	BIN3	39	237738
8:22288198	BIN3	39	237782
8:22288098	BIN3	39	237882
6:32340871	HLA-DRB5	31	237901
6:30346760	TRIM40	30	238077
8:22287870	BIN3	39	238110
6:32340236	HLA-DRB5	31	238536
17:76186618	DNAH17	72	238862
8:22286994	BIN3	39	238986
1:232903701	SIPA1L2	6	239090
1:205483124	NUCKS1	4	240448
8:22285507	BIN3	39	240473
8:22285490	BIN3	39	240490
8:22284603	BIN3	39	241377
8:22284217	BIN3	39	241763
8:22283989	BIN3	39	241991

snp_id	gwas_nearest_gene	gwas_locus_number	gwas_dist_bp
8:22283514	BIN3	39	242466
6:30351348	TRIM40	30	242665
8:22282748	BIN3	39	243232
8:22282732	BIN3	39	243248
8:22282416	BIN3	39	243564
8:22282190	BIN3	39	243790
12:132819486	FBRSL1	52	244282
6:30354303	TRIM40	30	245620
6:30357294	TRIM40	30	248611

Step 5: Union table — cases supported by metQTL, GWAS, or both

```
metqtl_snp_set <- unique(metqtl_cases_sig$SNP)

union_table <- cases_meta %>%
  mutate(in_metqtl = snp_id %in% metqtl_snp_set) %>%
  left_join(gwas_matches %>% select(snp_id, is_gwas_hit, gwas_dist_bp,
                                    gwas_nearest_gene, gwas_locus_number),
            by = "snp_id") %>%
  rename(in_gwas = is_gwas_hit) %>%
  left_join(overlap_cases %>% distinct(snp_id, metabolite_feature),
            by = "snp_id", relationship = "many-to-many") %>%
  # Union criteria: metQTL support, GWAS proximity, OR longitudinal dynamic
  # classification — a dynamic hit is of interest in its own right even
  # without external metQTL/GWAS support, per the earlier left-join finding
  # that 2 dynamic case hits had no metQTL overlap.
  mutate(union_hit = in_metqtl | in_gwas | dynamic) %>%
  filter(union_hit)

write_csv(union_table, file.path(RESULTS_DIR, "cases_union_metqtl_or_gwas.csv"))

cat("Total case pairs in union (metQTL, GWAS, and/or dynamic):", nrow(union_table), "\n")
```

```
## Total case pairs in union (metQTL, GWAS, and/or dynamic): 1134
```

```
cat("Unique SNPs in union:", n_distinct(union_table$snp_id), "\n")
```

```
## Unique SNPs in union: 938
```

```
cat("  - metQTL only:            ", n_distinct(union_table$snp_id[union_table$in_metqtl & !union_table$in_gwas & !
union_table$dynamic]), "\n")
```

```
##   - metQTL only:             32
```

```
cat("  - GWAS only:              ", n_distinct(union_table$snp_id[!union_table$in_metqtl & union_table$in_gwas & !
union_table$dynamic]), "\n")
```

```
##   - GWAS only:               905
```

```
cat("  - Dynamic only (neither):  ", n_distinct(union_table$snp_id[!union_table$in_metqtl & !union_table$in_gwas &
union_table$dynamic]), "\n")
```

```
##   - Dynamic only (neither):   1
```

```
cat("  - metQTL + GWAS:          ", n_distinct(union_table$snp_id[union_table$in_metqtl & union_table$in_gwas & !u
nion_table$dynamic]), "\n")
```

```
##   - metQTL + GWAS:           0
```

```
cat("  - Any combo incl. dynamic:", n_distinct(union_table$snp_id[union_table$dynamic & (union_table$in_metqtl | u
nion_table$in_gwas)]), "\n")
```

```
##   - Any combo incl. dynamic: 0
```

```
union_table %>%
  distinct(snp_id, cpg_id, category, in_metqtl, in_gwas, dynamic,
           metabolite_feature, gwas_nearest_gene, gwas_dist_bp) %>%
  arrange(desc(dynamic), desc(in_metqtl & in_gwas)) %>%
  kable(caption = "Case meta-hits supported by metQTL overlap, GWAS proximity, and/or longitudinal dynamic classif
ication")
```

Case meta-hits supported by metQTL overlap, GWAS proximity, and/or longitudinal dynamic classification

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
4:184324768	cg27037990	replicated_and_dynamic	FALSE	FALSE	TRUE	NA	CLCN3	13741611
6:112168309	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	74982
6:112164441	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	78850
6:112164313	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	78978
6:112161786	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	81505
6:112157615	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	85676
6:112149373	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	93918
6:112147107	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	96184
6:112148293	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	94998
6:112141908	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	101383
6:112141732	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	101559
6:112138294	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	104997
6:112131578	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	111713
6:112130478	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	112813
6:112128947	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	114344
6:112125335	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	117956
6:112124440	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	118851
6:112124438	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	118853
6:112122226	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	121065
6:112121766	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	121525
6:112121096	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	122195
6:112119619	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	123672
6:112119981	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	123310
6:112114791	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	128500
6:112114249	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	129042
6:112114308	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	128983
6:112113938	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	129353
6:112118505	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	124786
6:112121228	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	122063

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
6:112116451	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	126840
6:112115355	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	127936
6:112132032	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	111259
6:112111394	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	131897
6:112110528	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	132763
6:112151444	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	91847
6:112149373	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	93918
6:112147107	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	96184
6:112148293	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	94998
6:112138882	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	104409
6:112141908	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	101383
6:112141732	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	101559
6:112138294	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	104997
6:112131578	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	111713
6:112130478	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	112813
6:112128947	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	114344
6:112125335	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	117956
6:112124440	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	118851
6:112124438	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	118853
6:112114249	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	129042
6:112114308	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	128983
6:112114791	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	128500
6:112115355	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	127936
6:112121766	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	121525
6:112121096	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	122195
6:112119619	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	123672
6:112119981	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	123310
6:112113938	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	129353
6:112118505	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	124786
6:112116451	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	126840
6:112122226	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	121065
6:112121228	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	122063
6:112132032	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	111259
6:112111394	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	131897
6:112110528	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	132763
6:112125453	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	117838
6:112151444	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	91847
6:30037908	cg23363602	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	70775
6:112155473	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	87818
6:112154021	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	89270
6:112154200	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	89091
6:112154244	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	89047
6:112152494	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	90797

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
6:112152415	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	90876
6:112152768	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	90523
6:112152560	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	90731
6:112155473	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	87818
6:112154021	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	89270
6:112154200	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	89091
6:112154244	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	89047
6:112152768	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	90523
6:112152494	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	90797
6:112152415	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	90876
6:112152560	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	90731
2:169117820	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	7426
2:169118135	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	7741
2:169115051	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	4657
2:169120704	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	10310
2:169117706	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	7312
2:169118415	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	8021
2:169117008	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	6614
2:169119257	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	8863
2:169119317	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	8923
2:169118454	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	8060
2:169109956	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	438
2:169122942	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	12548
6:112131648	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	111643
2:169113808	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	3414
2:169114387	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	3993
2:169116027	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	5633
2:169122597	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	12203
2:169121700	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	11306
2:169121841	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	11447
2:169116312	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	5918
2:169131721	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	21327
12:133028325	cg17566325	replicated_static	FALSE	TRUE	FALSE	NA	FBRSL1	35443
2:169133371	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	22977
2:169129878	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	19484
2:169107532	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	2862
2:169109230	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	1164
2:169106850	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	3544
6:30043779	cg17968668	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	64904
2:169122644	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	12250
6:112151452	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	91839
2:169125462	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	15068
2:169125080	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	14686

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
2:169133002	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	22608
6:30046682	cg17968668	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	62001
6:30046429	cg17968668	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	62254
6:30046376	cg17968668	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	62307
6:30046005	cg17968668	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	62678
6:30046011	cg17968668	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	62672
6:30043955	cg17968668	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	64728
6:30044249	cg17968668	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	64434
2:169129601	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	19207
2:169102577	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	7817
17:7300611	cg10651007	replicated_static	FALSE	TRUE	FALSE	NA	CHRNB1	55010
6:112157615	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	85676
6:112149486	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	93805
6:112147909	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	95382
6:112168309	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	74982
6:112161786	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	81505
6:112164441	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	78850
6:112164313	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	78978
6:112143472	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	99819
6:112151444	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	91847
6:112129636	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	113655
6:112126303	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	116988
6:112171130	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	72161
6:112114151	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	129140
4:77172445	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	24476
4:77173739	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E-STBD1	24315
6:112115993	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	127298
2:169140909	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	30515
6:112113958	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	129333
6:112114791	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	128500
6:112114249	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	129042
6:112114308	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	128983
6:112111394	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	131897
6:112113938	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	129353
6:112110528	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	132763
6:112149373	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	93918
6:112147107	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	96184
6:112148293	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	94998
6:112138882	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	104409
6:112141732	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	101559
6:112141908	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	101383
6:112138294	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	104997
17:43933307	cg16228356	replicated_static	FALSE	TRUE	FALSE	NA	CRHR1	134999

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
6:112131578	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	111713
6:112130478	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	112813
6:112124440	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	118851
6:112119981	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	123310
6:112124438	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	118853
6:112128947	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	114344
6:112115355	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	127936
6:112125335	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	117956
6:112121766	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	121525
6:112119619	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	123672
6:112121096	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	122195
6:112116451	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	126840
6:112118505	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	124786
6:112122226	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	121065
6:112121228	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	122063
4:77171925	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	23956
4:77168321	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	20352
4:77168112	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	20143
13:49893105	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	34627
13:49889267	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	38465
13:49891933	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	35799
4:77151109	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	3140
4:77151110	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	3141
4:77151300	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	3331
4:77155139	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	7170
4:77154957	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	6988
4:77148761	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	792
4:77148601	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	632
13:49893024	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	34708
4:77165605	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	17636
4:77161299	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	13330
13:49894274	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	33458
4:77149772	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	1803
4:77150655	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	2686
4:77150838	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	2869
4:77145892	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	2077
4:77155486	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	7517
4:77149540	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	1571
4:77148213	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	244
6:112132032	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	111259
13:49890902	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	36830
13:49891231	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	36501
13:49886656	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	41076

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
4:77150468	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	2499
13:49894824	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	32908
13:49886855	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	40877
13:49943607	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	15875
5:125963051	cg08702413	replicated_static	TRUE	FALSE	FALSE	C18_mz_rt_518.25_185.639	C5orf24	8236054
5:125963303	cg08702413	replicated_static	TRUE	FALSE	FALSE	C18_mz_rt_518.25_185.639	C5orf24	8235802
4:77149119	cg01520956	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	1150
13:49878714	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	49018
13:49895644	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	32088
13:49878909	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	48823
13:49876171	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	51561
13:49876659	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	51073
2:169051157	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	59237
13:49866322	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	61410
13:49897778	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	29954
13:49897226	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	30506
13:49878236	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	49496
2:169069717	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	40677
2:169068506	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	41888
13:49883444	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	44288
13:49850649	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	77083
13:49850894	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	76838
13:49852430	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	75302
2:169060668	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	49726
13:49877951	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	49781
13:49878804	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	48928
13:49901088	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	26644
13:49877403	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	50329
13:49884055	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	43677
13:49848833	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	78899
2:169043083	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	67311
13:49864106	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	63626
6:112172746	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	70545
6:112176239	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	67052
13:49864107	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	63625
13:49867290	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	60442
13:49868341	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	59391
13:49864372	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	63360
13:49863811	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	63921
13:49869578	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	58154
6:112169138	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	74153
13:49873894	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	53838
13:49930369	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	2637

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
13:49841499	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	86233
13:49856198	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	71534
2:169036789	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	73605
13:49869445	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	58287
13:49838972	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	88760
13:49845485	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	82247
13:49846230	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	81502
13:49848194	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	79538
13:49848794	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	78938
13:49852377	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	75355
13:49854515	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	73217
13:49852000	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	75732
13:49844085	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	83647
13:49855276	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	72456
13:49824356	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	103376
13:49825667	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	102065
13:49839979	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	87753
13:49825360	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	102372
13:49858731	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	69001
13:49861155	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	66577
13:49860691	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	67041
13:49839548	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	88184
13:49846920	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	80812
13:49847452	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	80280
13:49843964	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	83768
13:49839889	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	87843
13:49862027	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	65705
6:112151452	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	91839
13:49873678	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	54054
2:169035989	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	74405
4:77131294	cg13870520	replicated_static	FALSE	TRUE	FALSE	NA	FAM47E	16675
7:23482044	cg07048044	replicated_static	FALSE	TRUE	FALSE	NA	GPNMB	181995
13:49830629	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	97103
13:49834252	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	93480
13:49834463	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	93269
13:49833774	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	93958
13:49831703	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	96029
13:49837022	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	90710
2:169035673	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	74721
7:23483373	cg07048044	replicated_static	FALSE	TRUE	FALSE	NA	GPNMB	183324
6:30026236	cg16586538	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	82447
6:30034289	cg16586538	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	74394
6:30108751	cg08996678	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	68

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
6:30106493	cg08996678	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	2190
6:30106202	cg08996678	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	2481
6:30105391	cg08996678	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	3292
6:30108336	cg08996678	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	347
6:30109063	cg08996678	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	380
6:30107153	cg08996678	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	1530
6:30107309	cg08996678	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	1374
2:169020642	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	89752
2:169016263	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	94131
2:169016171	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	94223
7:23490230	cg07048044	replicated_static	FALSE	TRUE	FALSE	NA	GPNMB	190181
7:23485919	cg07048044	replicated_static	FALSE	TRUE	FALSE	NA	GPNMB	185870
13:49826508	cg19238349	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	101224
5:60088148	cg20565082	replicated_static	FALSE	TRUE	FALSE	NA	ELOVL7	49811
2:169036091	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	74303
2:169126764	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	16370
2:169027555	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	82839
6:30064702	cg16586538	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	43981
14:88424120	cg01004063	replicated_static	FALSE	TRUE	FALSE	NA	GALC	40144
6:29937555	cg27020253	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	171128
6:30319154	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	210471
12:132819486	cg02410480	replicated_static	FALSE	TRUE	FALSE	NA	FBRSL1	244282
17:7245674	cg07148458	replicated_static	FALSE	TRUE	FALSE	NA	CHRNB1	109947
6:30068195	cg16586538	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	40488
14:88417821	cg01004063	replicated_static	FALSE	TRUE	FALSE	NA	GALC	46443
6:30335350	cg16206678	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	226667
14:88433660	cg01004063	replicated_static	FALSE	TRUE	FALSE	NA	GALC	30604
17:44927282	cg25836567	replicated_static	FALSE	TRUE	FALSE	NA	WNT3	60477
2:169012955	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	97439
17:44931434	cg25836567	replicated_static	FALSE	TRUE	FALSE	NA	WNT3	64629
17:44928769	cg25836567	replicated_static	FALSE	TRUE	FALSE	NA	WNT3	61964
6:29960420	cg16586538	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	148263
17:44930345	cg25836567	replicated_static	FALSE	TRUE	FALSE	NA	WNT3	63540
17:44930363	cg25836567	replicated_static	FALSE	TRUE	FALSE	NA	WNT3	63558
6:29966726	cg16586538	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	141957
17:44927816	cg25836567	replicated_static	FALSE	TRUE	FALSE	NA	WNT3	61011
17:44928053	cg25836567	replicated_static	FALSE	TRUE	FALSE	NA	WNT3	61248
17:44927374	cg25836567	replicated_static	FALSE	TRUE	FALSE	NA	WNT3	60569
17:44926818	cg25836567	replicated_static	FALSE	TRUE	FALSE	NA	WNT3	60013
2:169089920	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	20474
6:30323393	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	214710
6:30318327	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	209644
6:30299245	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	190562

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
6:30331216	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	222533
6:30251391	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	142708
6:30330392	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	221709
6:30329926	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	221243
6:30248711	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	140028
6:30279130	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	170447
17:44922276	cg25836567	replicated_static	FALSE	TRUE	FALSE	NA	WNT3	55471
6:30327839	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	219156
6:30328192	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	219509
6:30328357	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	219674
6:133016151	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	194210
6:133016000	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	194361
6:133015662	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	194699
6:133015756	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	194605
6:133015951	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	194410
6:30329966	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	221283
6:30330071	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	221388
6:30330737	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	222054
6:30336663	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	227980
2:169090729	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	19665
6:30337141	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	228458
6:133022197	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	188164
6:133016337	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	194024
6:133022468	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	187893
6:133022761	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	187600
6:133023036	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	187325
6:133024980	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	185381
6:30357294	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	248611
17:7310473	cg10651007	replicated_static	FALSE	TRUE	FALSE	NA	CHRNB1	45148
6:133016300	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	194061
6:133016902	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	193459
6:133016967	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	193394
6:133016685	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	193676
6:133017042	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	193319
2:169103143	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	7251
6:133024183	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	186178
6:133025637	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	184724
6:133025684	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	184677
6:30341390	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	232707
6:133030492	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	179869
6:133030710	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	179651
6:133030711	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	179650
6:30333505	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	224822

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
6:133031111	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	179250
6:133031252	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	179109
6:30340528	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	231845
6:30351348	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	242665
6:32741894	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	163122
6:32741006	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	162234
6:32742531	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	163759
6:32741742	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	162970
6:30337974	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	229291
6:30285312	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	176629
19:2363319	cg00513288	replicated_static	FALSE	TRUE	FALSE	NA	SPPL2B	22272
6:133032759	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	177602
6:30224970	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	116287
6:30346760	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	238077
6:32752788	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	174016
6:32750031	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	171259
6:32750279	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	171507
4:712222	cg06672737	replicated_static	FALSE	TRUE	FALSE	NA	GAK	213154
6:32753313	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	174541
6:32753682	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	174910
6:32753739	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	174967
6:32754094	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	175322
6:32749704	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	170932
6:32746984	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	168212
6:32747012	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	168240
6:32746845	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	168073
6:32747060	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	168288
6:32747273	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	168501
6:32753492	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	174720
6:32747959	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	169187
6:32747971	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	169199
6:30327194	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	218511
6:30326804	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	218121
6:32748633	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	169861
6:30326809	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	218126
6:32746920	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	168148
6:32755937	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	177165
6:32756363	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	177591
6:32752577	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	173805
6:32756874	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	178102
6:32756872	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	178100
6:32755216	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	176444
6:32756961	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	178189

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
6:32752566	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	173794
6:32754920	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	176148
17:78690019	cg15096353	replicated_static	TRUE	FALSE	FALSE	C18_mz_rt_385.311_268.674	DNAH17	2264539
6:32745156	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	166384
6:32745324	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	166552
6:32746012	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	167240
6:32757131	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	178359
6:32757396	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	178624
6:32757938	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	179166
6:133035325	cg16100392	replicated_static	FALSE	TRUE	FALSE	NA	RPS12	175036
6:32758966	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	180194
17:40514201	cg20291162	replicated_static	FALSE	TRUE	FALSE	NA	RETREG3	226812
6:32757158	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	178386
6:32757453	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	178681
6:32744684	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	165912
6:32744730	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	165958
6:30327196	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	218513
6:30114402	cg09196959	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	5719
6:112154021	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	89270
6:112154200	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	89091
6:112154244	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	89047
6:32755023	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	176251
6:112155473	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	87818
6:112152494	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	90797
6:112152415	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	90876
6:30037686	cg17608587	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	70997
1:155151754	cg01019262	replicated_static	FALSE	TRUE	FALSE	NA	KRTCAP2	16718
6:30044827	cg17608587	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	63856
6:112152560	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	90731
6:30043992	cg17608587	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	64691
6:32750982	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	172210
6:112152768	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	90523
6:112149486	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	93805
6:112147909	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	95382
6:32744667	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	165895
6:32744566	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	165794
2:169001183	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	109211
2:169000293	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	110101
17:40527544	cg20291162	replicated_static	FALSE	TRUE	FALSE	NA	RETREG3	213469
6:32744050	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	165278
6:32744321	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	165549
6:32744452	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	165680
4:743125	cg06672737	replicated_static	FALSE	TRUE	FALSE	NA	GAK	182251

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
6:32743861	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	165089
6:32743173	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	164401
6:32743193	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	164421
6:32744145	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	165373
6:32743217	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	164445
1:155151493	cg01019262	replicated_static	FALSE	TRUE	FALSE	NA	KRTCAP2	16457
6:112143472	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	99819
17:40525098	cg20291162	replicated_static	FALSE	TRUE	FALSE	NA	RETREG3	215915
17:40526273	cg20291162	replicated_static	FALSE	TRUE	FALSE	NA	RETREG3	214740
6:32742059	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	163287
6:30073776	cg01692379	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	34907
6:30073847	cg00720829	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	34836
1:155155731	cg01019262	replicated_static	FALSE	TRUE	FALSE	NA	KRTCAP2	20695
6:30354303	cg16206678	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	245620
12:133178417	cg19008320	replicated_static	FALSE	TRUE	FALSE	NA	FBRSL1	114649
1:155157635	cg01019262	replicated_static	FALSE	TRUE	FALSE	NA	KRTCAP2	22599
6:30073776	cg06486622	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	34907
2:169165476	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	55082
6:112129636	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	113655
4:704602	cg06672737	replicated_static	FALSE	TRUE	FALSE	NA	GAK	220774
6:112126303	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	116988
1:155169355	cg01019262	replicated_static	FALSE	TRUE	FALSE	NA	KRTCAP2	34319
1:155168930	cg01019262	replicated_static	FALSE	TRUE	FALSE	NA	KRTCAP2	33894
1:155157715	cg01019262	replicated_static	FALSE	TRUE	FALSE	NA	KRTCAP2	22679
5:60012521	cg20565082	replicated_static	FALSE	TRUE	FALSE	NA	ELOVL7	125438
6:30076163	cg05701418	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	32520
7:23475364	cg07048044	replicated_static	FALSE	TRUE	FALSE	NA	GPNMB	175315
6:112114151	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	129140
6:30128949	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	20266
6:30076064	cg05701418	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	32619
6:30127805	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	19122
6:32350384	cg15837308	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	228388
6:32375695	cg12672189	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	203077
6:30125543	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	16860
5:60024091	cg20565082	replicated_static	FALSE	TRUE	FALSE	NA	ELOVL7	113868
6:32762307	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	183535
6:32761826	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	183054
6:32762226	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	183454
6:32761776	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	183004
6:30075843	cg05701418	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	32840
6:32762103	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	183331
6:30167294	cg16206678	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	58611
6:30075641	cg05701418	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	33042

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
6:32761317	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	182545
6:32760965	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	182193
6:32761507	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	182735
4:1045824	cg26830958	replicated_static	FALSE	TRUE	FALSE	NA	TMEM175	93877
6:32760539	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	181767
2:169160724	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	50330
6:32765383	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	186611
6:32760257	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	181485
6:32760334	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	181562
6:32760249	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	181477
6:30076018	cg05701418	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	32665
6:32760655	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	181883
6:30074873	cg05701418	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	33810
6:32763974	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	185202
6:32763521	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	184749
6:32763821	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	185049
6:32764719	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	185947
6:32763040	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	184268
6:32763514	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	184742
6:32762616	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	183844
6:32762654	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	183882
6:32762484	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	183712
6:32767091	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	188319
6:32772004	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	193232
6:32771531	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	192759
6:32771354	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	192582
6:32771610	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	192838
6:32774385	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	195613
6:32771887	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	193115
6:32775465	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	196693
6:32775563	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	196791
6:32775602	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	196830
6:32775821	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	197049
6:32772636	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	193864
6:32777472	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	198700
6:32775686	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	196914
6:32774788	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	196016
6:32773042	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	194270
6:30076426	cg05701418	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	32257
6:32777216	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	198444
6:32777198	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	198426
6:32777533	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	198761
6:32776120	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	197348

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
6:32776414	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	197642
6:32776623	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	197851
6:32777305	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	198533
6:32776854	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	198082
6:32778153	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	199381
6:32778052	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	199280
6:32773172	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	194400
6:32770390	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	191618
6:32776976	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	198204
6:32777140	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	198368
6:32776832	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	198060
21:38891587	cg05434312	replicated_static	FALSE	TRUE	FALSE	NA	DYRK1A	39226
6:29942527	cg16586538	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	166156
6:30076016	cg05701418	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	32667
6:32367847	cg15837308	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	210925
6:32368087	cg15837308	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	210685
4:712048	cg23486923	replicated_static	FALSE	TRUE	FALSE	NA	GAK	213328
8:22289582	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	236398
6:112115993	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	127298
4:1046992	cg26830958	replicated_static	FALSE	TRUE	FALSE	NA	TMEM175	95045
2:169156742	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	46348
6:32778531	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	199759
8:22288259	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	237721
6:112113958	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	129333
2:177574919	cg12504882	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_562.633_68.487	STK39	8464525
8:22291265	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	234715
6:32627700	cg21893764	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	48928
6:32793770	cg23538064	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	214998
8:22288995	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	236985
6:29932897	cg01505556	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	175786
6:32780201	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	201429
6:30146019	cg16206678	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	37336
6:29934109	cg24245773	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	174574
7:23520104	cg07048044	replicated_static	FALSE	TRUE	FALSE	NA	GPNMB	220055
6:32780317	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	201545
6:72367279	cg06754764	replicated_static	FALSE	TRUE	FALSE	NA	RIMS1	120483
7:23517468	cg07048044	replicated_static	FALSE	TRUE	FALSE	NA	GPNMB	217419
6:30140342	cg16206678	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	31659
2:169163422	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	53028
6:32369355	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	209417
6:32369643	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	209129
6:32369886	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	208886
8:22289166	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	236814

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
8:22288789	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	237191
8:22288778	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	237202
8:22289093	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	236887
8:22289126	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	236854
8:22289288	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	236692
8:22288815	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	237165
8:22288889	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	237091
8:22288637	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	237343
8:22288242	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	237738
8:22288198	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	237782
8:22288342	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	237638
6:30179164	cg16206678	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	70481
6:30354303	cg05701418	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	245620
7:23519647	cg07048044	replicated_static	FALSE	TRUE	FALSE	NA	GPNMB	219598
4:790757	cg24922022	replicated_static	FALSE	TRUE	FALSE	NA	GAK	134619
6:32367604	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	211168
6:32361424	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	217348
8:22294836	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	231144
6:32367604	cg05973262	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	211168
4:790182	cg24922022	replicated_static	FALSE	TRUE	FALSE	NA	GAK	135194
6:32365284	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	213488
6:32364984	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	213788
6:29923351	cg07187855	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	185332
6:32367259	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	211513
8:22288098	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	237882
8:22287870	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	238110
7:23520963	cg07048044	replicated_static	FALSE	TRUE	FALSE	NA	GPNMB	220914
17:76323344	cg21940220	replicated_static	FALSE	TRUE	FALSE	NA	DNAH17	102136
15:61845512	cg08203682	replicated_static	FALSE	TRUE	FALSE	NA	VPS13C	151873
6:32365707	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	213065
8:22286994	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	238986
2:169107007	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	3387
6:32361485	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	217287
6:32361694	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	217078
6:32365707	cg05973262	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	213065
6:32361525	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	217247
6:32362930	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	215842
6:32361583	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	217189
8:11828200	cg16871763	replicated_static	FALSE	TRUE	FALSE	NA	CTSB	115757
6:32366421	cg09018483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	212351
6:32361003	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	217769
1:154796342	cg05746024	replicated_static	FALSE	TRUE	FALSE	NA	PMVK	101843
6:32361566	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	217206

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
6:32360714	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	218058
6:32362272	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	216500
6:32360641	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	218131
6:32360945	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	217827
6:32361130	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	217642
6:32361255	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	217517
6:32361223	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	217549
6:32360477	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	218295
10:100150148	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_162.132_101.515	GBF1	3865131
10:100150148	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_183.111_106.825	GBF1	3865131
10:100150148	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_161.128_100.935	GBF1	3865131
6:32363774	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	214998
6:32768980	cg12333338	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	190208
6:32360136	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	218636
6:32361894	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	216878
6:72300534	cg06754764	replicated_static	FALSE	TRUE	FALSE	NA	RIMS1	187228
6:32360069	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	218703
6:32360050	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	218722
6:32359800	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	218972
6:32360644	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	218128
6:32361251	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	217521
6:32360955	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	217817
6:32361080	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	217692
6:32360589	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	218183
7:23514868	cg07048044	replicated_static	FALSE	TRUE	FALSE	NA	GPNMB	214819
6:32366625	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	212147
6:32360970	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	217802
6:32367451	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	211321
6:32367927	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	210845
6:30199016	cg16206678	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	90333
6:32365896	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	212876
6:32365971	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	212801
6:32366331	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	212441
6:32359617	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	219155
17:42627902	cg13668490	replicated_static	FALSE	TRUE	FALSE	NA	FAM171A2	193272
6:32367515	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	211257
7:23521316	cg07048044	replicated_static	FALSE	TRUE	FALSE	NA	GPNMB	221267
6:29923351	cg26321999	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	185332
7:23519260	cg07048044	replicated_static	FALSE	TRUE	FALSE	NA	GPNMB	219211
7:23519710	cg07048044	replicated_static	FALSE	TRUE	FALSE	NA	GPNMB	219661
6:32359389	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	219383
6:29939677	cg08698936	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	169006
6:32362020	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	216752

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
6:32358834	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	219938
6:32357339	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	221433
6:32358345	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	220427
6:32350591	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	228181
6:32362010	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	216762
6:32358883	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	219889
6:32357333	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	221439
6:32358368	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	220404
6:32357489	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	221283
6:32358205	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	220567
6:32357525	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	221247
6:32357529	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	221243
6:32357397	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	221375
8:22285490	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	240490
6:32358113	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	220659
6:32357715	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	221057
4:1052009	cg26830958	replicated_static	FALSE	TRUE	FALSE	NA	TMEM175	100062
8:22282416	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	243564
6:32357037	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	221735
6:32356829	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	221943
6:32356356	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	222416
6:32354888	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	223884
6:32354520	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	224252
6:32355724	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	223048
6:32356876	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	221896
6:32355631	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	223141
6:32355981	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	222791
6:32356242	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	222530
6:32356440	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	222332
6:32356573	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	222199
6:32352294	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	226478
6:32356667	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	222105
6:32356489	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	222283
8:22282732	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	243248
6:32357133	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	221639
6:32357023	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	221749
6:32356272	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	222500
6:32356097	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	222675
6:32355109	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	223663
6:32356917	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	221855
10:100147060	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_162.132_101.515	GBF1	3868219
10:100147060	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_183.111_106.825	GBF1	3868219
10:100147060	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_161.128_100.935	GBF1	3868219

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
8:22282190	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	243790
8:22284603	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	241377
1:205483124	cg06948937	replicated_static	FALSE	TRUE	FALSE	NA	NUCKS1	240448
6:29923351	cg09965419	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	185332
10:100148653	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_162.132_101.515	GBF1	3866626
10:100148653	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_183.111_106.825	GBF1	3866626
10:100148653	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_161.128_100.935	GBF1	3866626
10:100146895	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_162.132_101.515	GBF1	3868384
10:100146895	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_183.111_106.825	GBF1	3868384
10:100146895	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_161.128_100.935	GBF1	3868384
10:100148004	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_162.132_101.515	GBF1	3867275
10:100148004	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_183.111_106.825	GBF1	3867275
10:100148004	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_161.128_100.935	GBF1	3867275
6:30120912	cg08996678	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	12229
10:100148353	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_162.132_101.515	GBF1	3866926
10:100148353	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_183.111_106.825	GBF1	3866926
10:100148353	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_161.128_100.935	GBF1	3866926
6:32361953	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	216819
6:32361943	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	216829
6:32361762	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	217010
6:32369643	cg09018483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	209129
6:32369355	cg09018483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	209417
6:32369886	cg09018483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	208886
10:100147813	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_162.132_101.515	GBF1	3867466
10:100147813	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_183.111_106.825	GBF1	3867466
10:100147813	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_161.128_100.935	GBF1	3867466
10:100148951	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_162.132_101.515	GBF1	3866328
10:100148951	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_183.111_106.825	GBF1	3866328
10:100148951	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_161.128_100.935	GBF1	3866328
6:32362745	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	216027
6:32362832	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	215940
6:32362741	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	216031
6:32362639	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	216133
6:32362669	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	216103
6:32362453	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	216319
10:100149346	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_162.132_101.515	GBF1	3865933
10:100149346	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_183.111_106.825	GBF1	3865933
10:100149346	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_161.128_100.935	GBF1	3865933
6:32362128	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	216644
6:32361388	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	217384
6:32362182	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	216590
2:1816813	cg22060041	replicated_static	TRUE	FALSE	FALSE	C18_mz_rt_142.944_65.269	KCNS3	16331035
2:1816762	cg22060041	replicated_static	TRUE	FALSE	FALSE	C18_mz_rt_142.944_65.269	KCNS3	16331086

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
6:32362075	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	216697
6:32361035	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	217737
6:32360551	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	218221
6:32360532	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	218240
6:32360443	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	218329
6:32360341	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	218431
6:32362217	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	216555
6:32360135	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	218637
6:32362262	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	216510
8:11830150	cg16871763	replicated_static	FALSE	TRUE	FALSE	NA	CTSB	117707
6:32362389	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	216383
6:32360060	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	218712
6:32362085	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	216687
10:100151189	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_162.132_101.515	GBF1	3864090
10:100151189	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_183.111_106.825	GBF1	3864090
10:100151189	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_161.128_100.935	GBF1	3864090
6:32375695	cg15837308	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	203077
10:100151799	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_162.132_101.515	GBF1	3863480
10:100151799	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_183.111_106.825	GBF1	3863480
10:100151799	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_161.128_100.935	GBF1	3863480
6:32359431	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	219341
10:100151844	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_162.132_101.515	GBF1	3863435
10:100151844	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_183.111_106.825	GBF1	3863435
10:100151844	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_161.128_100.935	GBF1	3863435
10:100152373	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_162.132_101.515	GBF1	3862906
10:100152373	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_183.111_106.825	GBF1	3862906
10:100152373	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_161.128_100.935	GBF1	3862906
6:32357344	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	221428
10:100143193	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_162.132_101.515	GBF1	3872086
10:100143193	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_183.111_106.825	GBF1	3872086
10:100143193	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_161.128_100.935	GBF1	3872086
6:32358728	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	220044
6:32359133	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	219639
6:32364984	cg09018483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	213788
6:32365284	cg09018483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	213488
6:32358008	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	220764
6:32358011	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	220761
17:44994859	cg07993743	replicated_static	FALSE	TRUE	FALSE	NA	WNT3	128054
6:32358213	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	220559
6:32358533	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	220239
6:32367259	cg09018483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	211513
6:32358459	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	220313
6:32343743	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	235029

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
6:32356839	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	221933
6:32354834	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	223938
6:32354629	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	224143
6:32355003	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	223769
6:32355677	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	223095
6:32355285	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	223487
6:32355303	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	223469
6:32356755	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	222017
6:32355858	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	222914
6:32356307	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	222465
6:32724043	cg07601645	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	145271
17:40653534	cg07471572	replicated_static	FALSE	TRUE	FALSE	NA	RETREG3	87479
17:40657101	cg07471572	replicated_static	FALSE	TRUE	FALSE	NA	RETREG3	83912
6:32785515	cg23538064	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	206743
6:32345131	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	233641
17:40650805	cg07471572	replicated_static	FALSE	TRUE	FALSE	NA	RETREG3	90208
17:7492388	cg10277912	replicated_static	FALSE	TRUE	FALSE	NA	CHRNB1	136767
17:7478402	cg10277912	replicated_static	FALSE	TRUE	FALSE	NA	CHRNB1	122781
8:22285507	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	240473
17:42629737	cg13668490	replicated_static	FALSE	TRUE	FALSE	NA	FAM171A2	195107
6:32359460	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	219312
17:76186618	cg15010903	replicated_static	FALSE	TRUE	FALSE	NA	DNAH17	238862
12:133152840	cg12011876	replicated_static	FALSE	TRUE	FALSE	NA	FBRSL1	89072
6:32790940	cg23538064	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	212168
6:32736079	cg07601645	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	157307
2:18077007	cg04112417	replicated_static	FALSE	TRUE	FALSE	NA	KCNS3	70841
10:100148176	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_162.132_101.515	GBF1	3867103
10:100148176	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_161.128_100.935	GBF1	3867103
10:100148176	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_183.111_106.825	GBF1	3867103
10:100148058	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_162.132_101.515	GBF1	3867221
10:100148058	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_161.128_100.935	GBF1	3867221
10:100148058	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_183.111_106.825	GBF1	3867221
8:22283514	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	242466
10:100141027	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_162.132_101.515	GBF1	3874252
10:100141027	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_183.111_106.825	GBF1	3874252
10:100141027	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_161.128_100.935	GBF1	3874252
6:32718345	cg07601645	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	139573
8:22283989	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	241991
8:22282748	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	243232
6:32359904	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	218868
8:22284217	cg17142470	replicated_static	FALSE	TRUE	FALSE	NA	BIN3	241763
6:112161092	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	82199
6:30094113	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	14570

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
6:32372963	cg27597473	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	205809
6:32372635	cg27597473	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	206137
6:32372863	cg27597473	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	205909
6:32373829	cg27597473	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	204943
6:32442086	cg25962829	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	136686
6:32372236	cg27597473	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	206536
6:32740671	cg05592483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	161899
6:32373185	cg27597473	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	205587
2:18083460	cg04112417	replicated_static	FALSE	TRUE	FALSE	NA	KCNS3	64388
6:32358144	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	220628
6:32358163	cg25317746	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	220609
6:30116078	cg08698936	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	7395
6:32600404	cg09018483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	21632
6:32600448	cg09018483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	21676
6:32600317	cg09018483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	21545
6:30096772	cg08698936	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	11911
6:32599999	cg09018483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	21227
10:100152055	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_162.132_101.515	GBF1	3863224
10:100152055	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_183.111_106.825	GBF1	3863224
10:100152055	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_161.128_100.935	GBF1	3863224
10:100151266	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_162.132_101.515	GBF1	3864013
10:100151266	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_183.111_106.825	GBF1	3864013
10:100151266	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_161.128_100.935	GBF1	3864013
10:100151305	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_162.132_101.515	GBF1	3863974
10:100151305	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_183.111_106.825	GBF1	3863974
10:100151305	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_161.128_100.935	GBF1	3863974
1:161684890	cg17036419	replicated_static	FALSE	TRUE	FALSE	NA	FCGR2A	215836
6:30084446	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	24237
6:32698391	cg23538064	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	119619
6:32602238	cg09018483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	23466
6:32603098	cg09018483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	24326
6:30095202	cg08698936	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	13481
6:30094315	cg08698936	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	14368
10:100142068	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_162.132_101.515	GBF1	3873211
10:100142068	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_183.111_106.825	GBF1	3873211
10:100142068	cg25079847	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_161.128_100.935	GBF1	3873211
12:133181425	cg19008320	replicated_static	FALSE	TRUE	FALSE	NA	FBRSL1	117657
6:29909988	cg08698936	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	198695
12:133156187	cg12011876	replicated_static	FALSE	TRUE	FALSE	NA	FBRSL1	92419
6:32427748	cg26757097	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	151024
2:169071190	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	39204
6:32340236	cg09737870	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	238536
1:161684890	cg18200075	replicated_static	FALSE	TRUE	FALSE	NA	FCGR2A	215836

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
12:133181399	cg19008320	replicated_static	FALSE	TRUE	FALSE	NA	FBRSL1	117631
6:32570401	cg09423875	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	8371
2:169065204	cg21899461	replicated_static	FALSE	TRUE	FALSE	NA	STK39	45190
6:32599485	cg09018483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	20713
9:136151806	cg21202204	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_530.207_109.904	UBAP2	102105415
6:29977869	cg20516460	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	130814
6:29983298	cg20516460	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	125385
2:18089614	cg04112417	replicated_static	FALSE	TRUE	FALSE	NA	KCNS3	58234
6:32349557	cg23224191	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	229215
6:32626537	cg15900054	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	47765
6:32343236	cg23224191	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	235536
6:32343339	cg23224191	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	235433
6:32367604	cg09018483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	211168
9:136154867	cg21202204	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_530.207_109.904	UBAP2	102108476
9:136154168	cg21202204	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_530.207_109.904	UBAP2	102107777
9:136153875	cg21202204	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_530.207_109.904	UBAP2	102107484
9:136154304	cg21202204	replicated_static	TRUE	FALSE	FALSE	HILIC_mz_rt_530.207_109.904	UBAP2	102107913
6:32706801	cg12296550	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	128029
6:32734304	cg12273368	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	155532
12:133282304	cg19512961	replicated_static	FALSE	TRUE	FALSE	NA	FBRSL1	218536
6:32703407	cg12296550	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	124635
6:30084696	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	23987
17:7245674	cg24664346	replicated_static	FALSE	TRUE	FALSE	NA	CHRNB1	109947
6:30084549	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	24134
6:30084505	cg26858414	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	24178
6:32591609	cg09018483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	12837
6:112157097	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	86194
6:112156463	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	86828
6:112156570	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	86721
6:112155722	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	87569
6:112155789	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	87502
6:112172975	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	70316
6:112175517	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	67774
6:30333505	cg13808979	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	224822
6:30336663	cg13808979	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	227980
6:30337141	cg13808979	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	228458
6:30323393	cg13808979	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	214710
6:30331216	cg13808979	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	222533
6:30328357	cg13808979	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	219674
6:30330392	cg13808979	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	221709
6:30329926	cg13808979	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	221243
6:30328192	cg13808979	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	219509
6:30327839	cg13808979	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	219156

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
6:30329966	cg13808979	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	221283
6:30330071	cg13808979	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	221388
6:30330737	cg13808979	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	222054
6:32685865	cg12296550	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	107093
6:30341390	cg13808979	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	232707
6:30340528	cg13808979	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	231845
6:30357294	cg13808979	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	248611
6:30351348	cg13808979	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	242665
6:32600914	cg09018483	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	22142
6:112250110	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	6819
4:15541609	cg16509355	replicated_static	FALSE	TRUE	FALSE	NA	BST1	195739
6:112131648	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	111643
6:112243291	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	0
6:112161092	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	82199
6:112155722	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	87569
6:112156570	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	86721
6:112156463	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	86828
6:112157097	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	86194
6:112155789	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	87502
6:112242253	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	1038
6:112165576	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	77715
6:112192795	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	50496
6:112194496	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	48795
6:112272725	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	29434
6:112200021	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	43270
6:112200061	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	43230
11:133718740	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	68261
6:112175735	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	67556
6:112203750	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	39541
6:112203035	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	40256
6:112208191	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	35100
11:133734177	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	52824
11:133720077	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	66924
11:133712348	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	74653
11:133712473	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	74528
11:133713189	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	73812
11:133712594	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	74407
11:133719565	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	67436
6:112267233	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	23942
6:112266347	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	23056
11:133727535	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	59466
11:133713730	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	73271
11:133714522	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	72479

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
11:133734322	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	52679
11:133713094	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	73907
11:133713082	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	73919
11:133731644	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	55357
6:112172975	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	70316
11:133714672	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	72329
6:112175517	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	67774
11:133715791	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	71210
11:133713436	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	73565
11:133733893	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	53108
11:133718467	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	68534
11:133717902	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	69099
19:2432781	cg13496660	replicated_static	FALSE	TRUE	FALSE	NA	SPPL2B	91734
19:2432177	cg13496660	replicated_static	FALSE	TRUE	FALSE	NA	SPPL2B	91130
6:112208753	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	34538
6:29914478	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	194205
6:29914553	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	194130
6:29914544	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	194139
11:133715153	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	71848
6:112216413	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	26878
6:29913824	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	194859
6:112212449	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	30842
6:112211802	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	31489
6:112212934	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	30357
11:133733864	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	53137
6:112213313	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	29978
6:29913886	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	194797
6:29911545	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	197138
11:133732150	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	54851
6:29912938	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	195745
11:133719221	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	67780
6:29909999	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	198684
6:29911604	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	197079
6:29911571	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	197112
6:29913449	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	195234
6:29914064	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	194619
6:29908110	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	200573
11:133716356	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	70645
6:112215441	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	27850
6:29913830	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	194853
6:29912297	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	196386
6:29910947	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	197736
6:29916654	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	192029

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
6:112151452	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	91839
6:29916639	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	192044
19:2435334	cg13496660	replicated_static	FALSE	TRUE	FALSE	NA	SPPL2B	94287
6:112218167	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	25124
6:29915791	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	192892
6:112218538	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	24753
6:29915694	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	192989
6:29915438	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	193245
6:29916480	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	192203
6:112158134	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	85157
6:29908892	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	199791
6:112221611	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	21680
6:29921999	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	186684
6:29915839	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	192844
6:29910402	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	198281
6:29906489	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	202194
6:29921460	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	187223
1:232903701	cg09548403	replicated_static	FALSE	TRUE	FALSE	NA	SIPA1L2	239090
6:29915351	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	193332
6:112153422	cg26846943	replicated_static	FALSE	TRUE	FALSE	NA	FYN	89869
6:29912281	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	196402
11:133726067	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	60934
6:29912713	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	195970
11:133725989	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	61012
19:2436424	cg13496660	replicated_static	FALSE	TRUE	FALSE	NA	SPPL2B	95377
19:2437058	cg13496660	replicated_static	FALSE	TRUE	FALSE	NA	SPPL2B	96011
6:29913698	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	194985
19:2439186	cg13496660	replicated_static	FALSE	TRUE	FALSE	NA	SPPL2B	98139
19:2439215	cg13496660	replicated_static	FALSE	TRUE	FALSE	NA	SPPL2B	98168
19:2439163	cg13496660	replicated_static	FALSE	TRUE	FALSE	NA	SPPL2B	98116
19:2439319	cg13496660	replicated_static	FALSE	TRUE	FALSE	NA	SPPL2B	98272
11:133729469	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	57532
11:133730820	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	56181
11:133735810	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	51191
11:133737304	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	49697
6:112113958	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	129333
6:112179226	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	64065
6:29916922	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	191761
6:29916971	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	191712
6:29919819	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	188864
11:133731134	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	55867
6:29922885	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	185798
6:29912386	cg22473097	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	196297

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
6:112210443	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	32848
6:112184621	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	58670
6:112184520	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	58771
6:112185838	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	57453
6:112185732	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	57559
6:112187553	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	55738
6:112182111	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	61180
6:112189712	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	53579
6:112189294	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	53997
6:32343369	cg12387154	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	235403
11:133732265	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	54736
11:133737462	cg15485101	replicated_static	FALSE	TRUE	FALSE	NA	IGSF9B	49539
6:112115993	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	127298
6:112150430	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	92861
6:112149486	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	93805
6:112147909	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	95382
6:112114151	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	129140
19:2435334	cg25966599	replicated_static	FALSE	TRUE	FALSE	NA	SPPL2B	94287
1:205653756	cg11965913	replicated_static	FALSE	TRUE	FALSE	NA	NUCKS1	69816
6:32358201	cg12387154	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	220571
6:112143472	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	99819
19:2432781	cg25966599	replicated_static	FALSE	TRUE	FALSE	NA	SPPL2B	91734
6:32345689	cg12387154	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	233083
6:32340871	cg12387154	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	237901
19:2432177	cg25966599	replicated_static	FALSE	TRUE	FALSE	NA	SPPL2B	91130
6:32368462	cg12387154	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	210310
6:112138882	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	104409
4:957284	cg20456258	replicated_static	FALSE	TRUE	FALSE	NA	TMEM175	5337
6:112165576	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	77715
6:112129636	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	113655
6:112158134	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	85157
6:112126303	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	116988
6:112153422	cg20596647	replicated_static	FALSE	TRUE	FALSE	NA	FYN	89869
19:2436424	cg25966599	replicated_static	FALSE	TRUE	FALSE	NA	SPPL2B	95377
19:2437058	cg25966599	replicated_static	FALSE	TRUE	FALSE	NA	SPPL2B	96011
6:32798548	cg01244342	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	219776
19:2439186	cg25966599	replicated_static	FALSE	TRUE	FALSE	NA	SPPL2B	98139
19:2439215	cg25966599	replicated_static	FALSE	TRUE	FALSE	NA	SPPL2B	98168
19:2439163	cg25966599	replicated_static	FALSE	TRUE	FALSE	NA	SPPL2B	98116
19:2439319	cg25966599	replicated_static	FALSE	TRUE	FALSE	NA	SPPL2B	98272
3:182913092	cg21794419	replicated_static	FALSE	TRUE	FALSE	NA	MCCC1	153019
6:32408497	cg12387154	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	170275
6:32387432	cg12387154	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	191340

snp_id	cpg_id	category	in_metqtl	in_gwas	dynamic	metabolite_feature	gwas_nearest_gene	gwas_dist_bp
6:32381280	cg12387154	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	197492
6:112125453	cg08601457	replicated_static	FALSE	TRUE	FALSE	NA	FYN	117838
6:32414435	cg12387154	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	164337
6:32395726	cg12387154	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	183046
3:182912282	cg21794419	replicated_static	FALSE	TRUE	FALSE	NA	MCCC1	152209
6:32424594	cg12387154	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	154178
13:50101526	cg10196532	replicated_static	FALSE	TRUE	FALSE	NA	CAB39L	173794
6:32431867	cg16609995	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	146905
6:32432509	cg16609995	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	146263
6:32431606	cg16609995	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	147166
6:32440467	cg16609995	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	138305
6:32440362	cg16609995	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	138410
6:32440910	cg16609995	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	137862
6:32441555	cg16609995	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	137217
6:32440879	cg16609995	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	137893
6:32435044	cg16609995	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	143728
6:32441408	cg16609995	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	137364
3:182912526	cg21794419	replicated_static	FALSE	TRUE	FALSE	NA	MCCC1	152453
6:32430289	cg16609995	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	148483
6:32429594	cg16609995	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	149178
6:32431147	cg16609995	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	147625
6:32439964	cg16609995	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	138808
6:32440451	cg16609995	replicated_static	FALSE	TRUE	FALSE	NA	HLA-DRB5	138321
6:30042861	cg24177217	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	65822
6:30038823	cg24177217	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	69860
6:30039767	cg24177217	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	68916
6:30041509	cg24177217	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	67174
6:30042057	cg24177217	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	66626
6:30032673	cg24177217	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	76010
6:30016297	cg24177217	replicated_static	FALSE	TRUE	FALSE	NA	TRIM40	92386

Step 6: Metabolite compound annotation (xmsannotator)

No collaborator compound-ID mapping table for the overlap metabolite features ever materialized, so this step resolves them independently: mass/retention-time matching (10ppm, ±30s) against the HMDB, KEGG, and LipidMaps reference databases via xmsannotator (https://github.com/kuppal2/xmsannotator), run separately per platform (`c18` / `hilic`) and database in `annotate_overlap_metabolites_xmsannotator.R`. `c18` / `LipidMaps` did not finish that run (only 2 initial DB matches, stopped short of a compound call); the other 5 platform/db combinations completed. `rerun_step3_matching.R` re-derives the consolidated match tables read in below from that run's on-disk output without re-running the (expensive) annotation itself.

```
best_annotation <- read_csv(file.path(RESULTS_DIR, "overlap_metabolite_best_annotation.csv"), show_col_types = FAL
SE) %>%
  select(metabolite_feature, annotated_compound, confidence, score, source)

n_any_match  <- n_distinct(best_annotation$metabolite_feature)
named_match  <- best_annotation %>% filter(confidence == 2, !annotated_compound %in% c("", "-"))

cat("Overlap metabolite features with any mz/rt match:      ", n_any_match, "/ 33\n")
```

```
## Overlap metabolite features with any mz/rt match:      23 / 33
```

```
cat("Overlap metabolite features with a high-confidence,\n",
    "named compound match:                                 ", nrow(named_match), "/ 33\n")
```

```
## Overlap metabolite features with a high-confidence,
##  named compound match:                                 11 / 33
```

```
overlap_cases_compounds <- overlap_cases_annot %>%
  left_join(best_annotation, by = "metabolite_feature")

overlap_controls_compounds <- overlap_controls_annot %>%
  left_join(best_annotation, by = "metabolite_feature")

union_table <- union_table %>%
  left_join(best_annotation, by = "metabolite_feature")

write_csv(overlap_cases_compounds, file.path(RESULTS_DIR, "mqtl_metqtl_overlap_cases_annotated_compounds.csv"))
write_csv(overlap_controls_compounds, file.path(RESULTS_DIR, "mqtl_metqtl_overlap_controls_annotated_compounds.cs
v"))
write_csv(union_table, file.path(RESULTS_DIR, "cases_union_metqtl_or_gwas.csv"))
```

Annotated compounds

A `confidence` of 2 / `score` of 100000 marks xmsannotator's high-confidence named calls; `confidence` 0 entries are a raw mass/RT hit only (frequently without a resolvable compound name) and are not treated as identifications below.

```
named_match %>%
  arrange(source, metabolite_feature) %>%
  select(metabolite_feature, annotated_compound, source) %>%
  kable(caption = "High-confidence (confidence = 2) compound matches for overlap metabolite features")
```

High-confidence (confidence = 2) compound matches for overlap metabolite features

metabolite_feature	annotated_compound	source
C18_mz_rt_180.067_36.514	N-Hydroxy-L-phenylalanine;2-(Hydroxyamino)-3-phenylpropanoate	c18_KEGG
C18_mz_rt_274.039_31.579	L-Tyrosine methyl ester 4-sulfate	c18_KEGG
C18_mz_rt_276.018_28.086	N-Phosphoguanidinoethyl methyl phosphate	c18_KEGG
HILIC_mz_rt_217.155_81.683	Valyl-Valine	hilic_HMDB
HILIC_mz_rt_253.143_145.88	QH(2)	hilic_HMDB
HILIC_mz_rt_278.037_62.69	()-Furilazole	hilic_HMDB
HILIC_mz_rt_161.128_100.935	N6-Methyl-L-lysine	hilic_KEGG
HILIC_mz_rt_353.343_289.84	111519-trimethyl-5Z-eicosenoic acid	hilic_LipidMaps
HILIC_mz_rt_409.313_37.218	4Z7Z10Z13Z16Z19Z22Z25Z-octacosaoctaenoic acid	hilic_LipidMaps
HILIC_mz_rt_412.307_39.743	N-(9Z-octadecenoyl)-glutamic acid	hilic_LipidMaps
HILIC_mz_rt_574.454_36.356	N-(tetradecanoyl)-deoxysphing-4-enine-1-sulfonate	hilic_LipidMaps

- The **GPX5/GPX6** metabolite feature (`C18_mz_rt_351.997_25.714`) got **no match** in either c18 table — it remains unresolved, so the glutathione/redox-pathway hypothesis raised earlier is still unconfirmed.
- The **ERLIN1** SNP's metabolite features fared better: two of its four associated features resolve to high-confidence matches, **N6-Methyl-L-lysine** and **Valyl-Valine** — both amino-acid derivatives, consistent with ERLIN1's role in ER lipid/protein handling but not a slam-dunk pathway link on their own.
- The **RUNX1** metabolite feature (`C18_mz_rt_709.527_218.804`) got no match.
- These are still single mass/RT hits within a fairly loose tolerance (10ppm/30s), not MS/MS-confirmed identities — treat names here as leads for literature/pathway follow-up, not confirmed compound IDs.
- The GWAS overlap (Step 4/5) is currently **cases only**, per the original request — extending the same window-match logic to the controls-side classified results is a straightforward addition if useful later.
- The `union_table` combines metQTL support, GWAS proximity, and longitudinal dynamic classification at the SNP level; a hit can qualify through any one criterion independently, and "any combo incl. dynamic" is the strongest converging evidence in this analysis — a locus independently

implicated by PD genetic risk, metabolomics, methylation, and a time-varying genetic effect.

- The 2 `replicated_and_dynamic` case-level hits are now included in the union table regardless of metQTL/GWAS support (Step 5), and are shown explicitly in Step 2's dynamic-status table whether or not they overlap a metQTL.
- Full output files (`*_combined_fdr.csv` , `*_sig_fdr05.csv` , `mqtl_metqtl_overlap_*_annotated.csv` , `overlap_snp_gene_annotations.csv` , `cases_union_metqtl_or_gwas.csv` , `mqtl_metqtl_overlap_*_annotated_compounds.csv`) are written alongside this report for downstream use. The `*_annotated.csv` files (Step 3) stay gene-annotation-only, since they're also the input contract for `annotate_overlap_metabolites_xmsannotator.R` / `rerun_step3_matching.R` ; the compound annotations from Step 6 are joined into separate `*_annotated_compounds.csv` files and into `cases_union_metqtl_or_gwas.csv` instead, to avoid changing that contract.